Mobile Broadband Upgrading and Spatial Inequality in South Africa

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Abstract

How do mobile broadband improvements differ across places with different initial quality and poverty levels? Faster connections can have economic value after basic access is established, making the distribution of service improvements relevant to broadband evaluation. We follow South Africa's 2021–2024 spectrum transition using quarterly Ookla measurements. Initially slow municipalities reduced their low-speed disadvantage and relative speed gap. Their absolute Mbps gap grew. ICASA-designated priority areas also reduced their low-speed disadvantage, alongside widening gaps in average speed and high-speed tile shares. A separate, exploratory comparison associates higher baseline poverty with less high-speed expansion; its precision varies with inference and measurement coverage. The findings distinguish progress within underserved places from changes in their position relative to other places. Evaluating upgrading therefore calls for tracking weak-service reduction, average performance, and high-speed expansion together, with measurement coverage reported for each comparison.

Keywords: mobile broadband; spectrum reform; spatial inequality; poverty; crowdsourced measurement; South Africa.

JEL classification: L96, L98, O18, O33.

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1 Introduction

A broadband upgrade can improve service in a poorly served municipality even as its speed gap with better-served places grows. Evaluating upgrading therefore requires examining both local improvements and differences across places. We ask how observed mobile broadband improvements differ across South African municipalities with different initial quality and poverty levels.

Faster connections can have economic value after basic internet access is established. Canzian, Poy, and Schüller (2019) study an ADSL2+ upgrade in rural Italy and estimate increases in firm revenue and total factor productivity. Fabling and Grimes (2021) find productivity gains from ultrafast broadband adoption in New Zealand, with clearer gains among firms making complementary organizational investments. Ahlfeldt, Koutroumpis, and Valletti (2017) show that faster connections are valued in English property markets, with diminishing returns to speed. These studies examine higher-performance availability or adoption. They make the distribution of subsequent quality improvements economically relevant, with returns that depend on use and local conditions.

Research on digital inequality has documented geographic and socioeconomic differences in connection quality. Gallardo and Whitacre (2024) relate fixed-broadband speed levels to socioeconomic characteristics in U.S. census tracts. Riddlesden and Singleton (2014) document English speed disparities over time, and Hilbert (2016) follows international bandwidth inequality. OECD (2025) reports rising speeds alongside widening absolute geographic gaps, including mobile networks. We extend these comparisons by following low-speed service, average performance, and high-speed expansion within one mobile transition. Linking these measures to a regulatory priority group shows how different assessments of upgrading can emerge for the same underserved places.

South Africa combines a major spectrum transition with explicit geographic rollout priorities. ICASA allocated additional spectrum in March 2022; licence validity began in July, and broadcasting relocation affected the later availability of low-frequency bands. The licensing framework prioritized severely under-serviced Batch 3 municipalities in coverage rollout. We use quarterly Ookla measurements to track improvements by initial service quality, baseline household poverty, and Batch 3 status. These classifications provide complementary perspectives: observed starting performance, socioeconomic disadvantage, and regulatory priority.

Average municipal download speed more than doubled between 2021Q1 and 2024Q4. Initially slow municipalities reduced their low-speed disadvantage and relative speed gap, alongside an increase in their absolute Mbps gap. Batch 3 municipalities show a related pattern: their disadvantage in very slow service narrowed, while their shortfalls in average speed and high-speed tile shares widened. The separate poverty comparison suggests less

high-speed expansion in poorer municipalities. Adding initial speed leaves the estimated poverty gradient similar in size, with statistical support sensitive to the inference procedure.

These findings show why improvement within underserved places and narrowing gaps between places deserve separate assessment. Following several performance measures reveals which disadvantages recede and which remain or grow during upgrading. This provides a concrete way to evaluate the geographic distribution of better service, including in areas explicitly prioritized by the regulator. Sections 2 and 3 describe the setting, data, and comparisons. Sections 4 and 5 present the results, followed by their economic and policy implications. The supplementary policy DDD and GDP-scaled light analyses are in Appendices B and E.

2 South Africa's Mobile Broadband Upgrading

2.1 Spectrum allocation and implementation

Spectrum assignment changes the conditions under which operators can expand capacity and coverage. Quantity, timing, and cost can matter for network quality (Bahia and Castells, 2022). South Africa's transition involved several dates rather than a single implementation event. Temporary spectrum was released in April 2020, followed by provisional assignments from December 1, 2021.[1]

The 2022 auction allocated spectrum in the 700, 800, 2600, and 3500 MHz bands. The opt-in stage took place on March 8, and the main stage ended on March 17. Licence validity began on July 1, 2022.[2] Low-frequency availability also depended on broadcasting relocation. The remaining analogue television transmissions in the 700 and 800 MHz bands were moved below 694 MHz by July 31, 2023.[3] Reallocation requires coordination with incumbent users, a broader issue in spectrum reform (Milgrom and Segal, 2020).

We follow the full 2021Q1–2024Q4 performance path. Period-average comparisons use 2021 and 2023–2024, setting aside the 2022 transition year. These comparisons describe upgrading during the transition. The separate institutional DDD in Appendix B examines earlier-access histories and their identifying assumptions.

[1] ICASA, April 2020, emergency spectrum; ICASA, March 28, 2022, provisional assignments and licence validity.

[2] ICASA, March 17, 2022, auction results; the July validity date is specified in ICASA's March 28 announcement. Licence issuance by May 7 is documented in paragraph 88 of the Constitutional Court judgment of June 28, 2022.

[3] ICASA, August 3, 2023, clearance of the 700/800 MHz bands.

2.2 Regulatory priority and existing disadvantage

Coverage obligations can direct provision toward places that commercial rollout has served poorly (Cave and Nicholls, 2017). ICASA's 2021 invitation to apply classifies municipalities using computer, cellphone, and internet access in the 2016 Community Survey, excluding landlines. Its Appendix D designates 99 of 213 municipalities as severely under-serviced Batch 3. Outside-in rollout provisions prioritize these places before Batch 2 and Batch 1.[4]

The balanced performance sample contains 92 Batch 3 and 114 other municipalities. Seven Batch 3 places lack a complete speed panel. Regulatory priority is distinct from measured initial speed: Batch 3 includes 35 of the 52 initially slowest municipalities and 12 of the 52 initially fastest. Poverty provides a third classification on a separately matched geography. Appendix Figure A1 locates the policy groups and initial performance.

3 Data and Empirical Comparisons

3.1 Observed service performance

We use the quarterly mobile layer of Speedtest by Ookla Global Fixed and Mobile Network Performance Maps. The layer aggregates tests from cellular connection types with sufficiently precise locations into spatial tiles.[5] Let v_{jt} denote a tile's mean download speed in Mbps and n_{jt} its test count. For observed tiles $\mathcal{J}_{it}$ in municipality i, mean speed is

$$M_{it} = \frac{\sum_{j \in \mathcal{J}_{it}} n_{jt} v_{jt}}{\sum_{j \in \mathcal{J}_{it}} n_{jt}}. \tag{1}$$

The mean-speed regression outcome is $\log M_{it}$. Within each municipality-quarter, the tile-speed 25th percentile and median give each observed tile equal weight. We also calculate

$$L_{it}(c) = \frac{1}{|\mathcal{J}_{it}|} \sum_{j \in \mathcal{J}_{it}} \mathrm{I}\{v_{jt} < c\}, \qquad H_{it}(50) = \frac{1}{|\mathcal{J}_{it}|} \sum_{j \in \mathcal{J}_{it}} \mathrm{I}\{v_{jt} \geq 50\}. \tag{2}$$

Low-speed shares use $c = 10$ and $c = 25$ Mbps; the higher-speed share uses 50 Mbps. These cutoffs describe parts of the observed speed distribution. They are not estimated thresholds for productive activity, household welfare, or licence compliance. Each share uses observed tiles as its denominator.

Table 1 distinguishes the geographic frames. The main performance sample contains 206 municipalities on 2018 boundaries, observed in all sixteen quarters. Initial groups are fixed

[4] ICASA, 2021 invitation to apply, Sections 12.2.1–12.2.2 and Appendix D.

[5] Ookla, data documentation.

using mean log download speed over observed 2019 quarters. Their sizes are 52, 51, 51, and 52. Of these places, 198 have four observed baseline quarters and eight have three. The separate baseline year avoids defining groups with the same observations used to calculate later changes; it does not remove baseline measurement error or regression to the mean.

Table 1. Analysis frames and comparisons

Analysis	Geography	Places	Comparison or role
Full performance frame	2018 boundaries	213	2021Q1–2024Q4; eight missing cells
Main performance and DDD	2018 boundaries	206	Complete 16-quarter panel; scalar regressions use 2021 and 2023–2024
Initial performance	Same main sample	206	2019 mean log speed; 198 have four baseline quarters and eight have three
Poverty gradient	2011 boundaries	223	Complete panel; 2016 poverty headcount and changes from 2021 to 2023–2024
BDM diagnostic	Late provinces	63	Separate broadcasting-exposure comparison in four provinces

Notes: The municipal universes contain 213 places on 2018 boundaries and 234 on 2011 boundaries. The main scalar policy panel contains $206 \times 12 = 2,472$ observations; its event study contains $206 \times 16 = 3,296$. Poverty scalar regressions contain one change observation for each of 223 municipalities.

The poverty analysis reaggregates tile measurements to 2011 municipal boundaries. It matches the 2016 Community Survey household multidimensional-poverty headcount, $H_{i,2016}$.[6] Of 234 municipalities in this geography, 223 are observed throughout 2021–2024. The headcount measures the share of households classified as multidimensionally poor. Data construction and the missing observations are documented in Appendix A.

[6]Statistics South Africa, Community Survey 2016: Provinces at a Glance.

3.2 Period changes and baseline comparisons

The basic regressions use one period change per municipality. We relate the change between the 2021 average and the 2023–2024 average to standardized 2019 initial speed, Z_i:

$$\Delta Y_i = a_{p(i)} + \rho Z_i + u_i, \qquad \Delta Y_i = \overline{Y}_{i,2023:2024} - \overline{Y}_{i,2021}. \tag{3}$$

Province effects $a_{p(i)}$ absorb common provincial changes. The coefficient ρ measures how subsequent changes differ with a one-standard-deviation higher initial speed. The quartile figure is a descriptive group comparison; this regression uses initial speed continuously.

The poverty comparison instead estimates

$$\Delta Y_i = c_{p(i)} + \theta(H_{i,2016}/10) + e_i. \tag{4}$$

One unit of $H_{i,2016}/10$ is ten percentage points of baseline poverty. Province effects $c_{p(i)}$ give a within-province change comparison. This baseline model relates changes to poverty within the poverty-compatible geography; the next specification adds initial speed.

As a robustness check, we add mean log download speed over observed 2019 quarters to Equation (4). The control is constructed on the same 2011 municipal boundaries. The 223 municipalities, outcome changes, province effects, and inference procedures are unchanged. Appendix C.1 reports the augmented specification and all six comparisons.

The main comparisons describe observed upgrading across municipalities. They relate changes to existing classifications and baseline conditions; they do not isolate effects of spectrum reform, regulatory priority, or poverty from other local changes. The institutional DDD and its additional identifying assumptions are presented separately in Appendix B.

3.3 Inference and observation support

Both tables report all six outcomes: mean, P25, and median speed in logs, and the three tile shares in percentage points. Period-change coefficients and full distribution summaries answer different questions (Quah, 1993). Standard errors cluster by province. With nine clusters, CR1 tests use $t(8)$; restricted wild-cluster tests enumerate all 512 Rademacher sign patterns (Cameron, Gelbach, and Miller, 2008; Cameron and Miller, 2015). Benjamini–Hochberg values adjust wild probabilities across the six outcomes within each analysis (Benjamini and Hochberg, 1995). Figures select comparisons for exposition; the six-outcome families govern the main multiplicity adjustments. Table 3 also retains the earlier two-threshold/four-other grouping, whose pre-results selection is not documented. None of these adjustments covers all analytical choices in the paper.

Crowdsourced tests observe selected devices, users, and locations. Measurement design therefore needs to match the question (Frías et al., 2025); representation can also differ between fixed and mobile tests (Bogulski et al., 2024). We compare all observed tiles with continuously measured locations and report support by poverty quartile. Section 5.3 reports the coverage and interpretation of these checks.

4 Where Did Mobile Performance Improve?

4.1 The scale and timing of aggregate upgrading

Average municipal download speed increased from 28.0 to 61.1 Mbps between 2021Q1 and 2024Q4 (Figure 1). The fixed 206-municipality series closely follows the all-available mean. The shaded area shows the cross-municipality interquartile range.

Performance improves before the July 2022 licence marker, rises later in 2022, falls in early 2023, and then resumes increasing. The path leaves room for deployment, congestion, devices, and measurement composition to change. Appendix Figure A6 shows upgrading in both the full tile distribution and the continuously observed tile sample. These aggregate paths establish the setting for the regional comparisons.

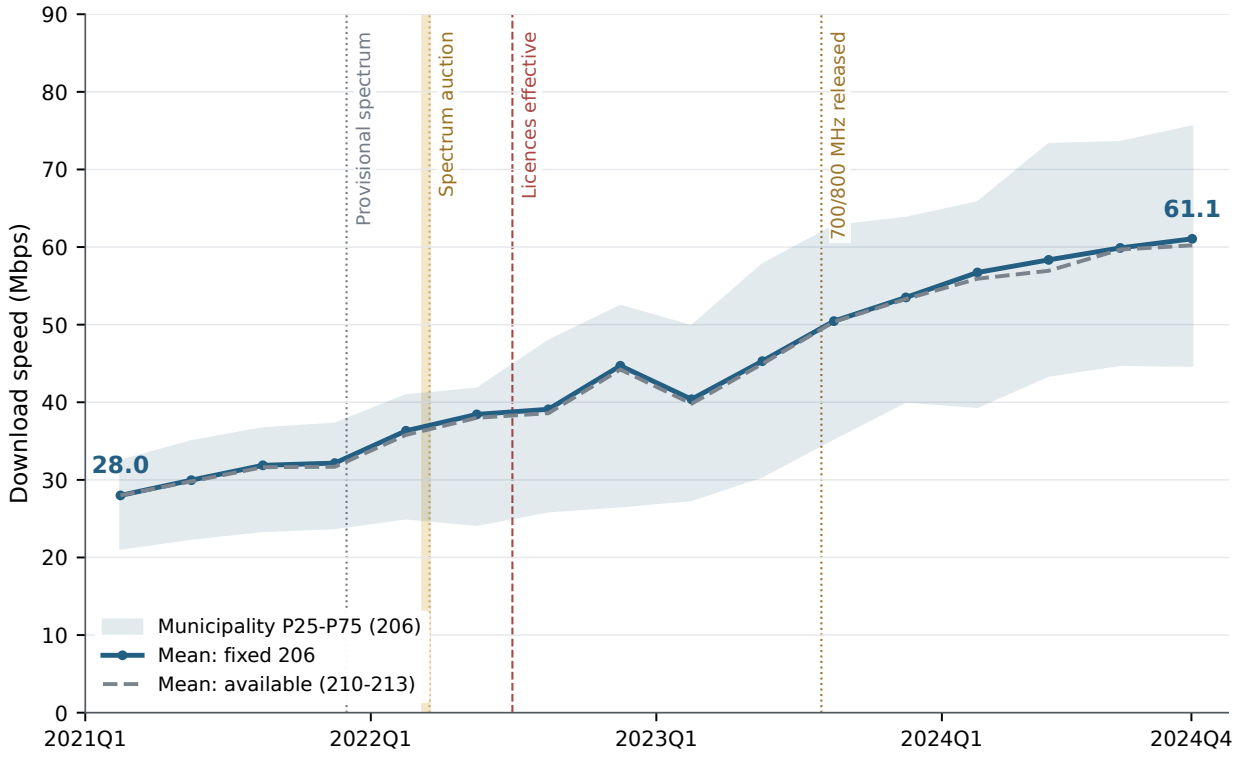


Figure 1. Municipal download performance during the spectrum transition

Notes: Each municipality-quarter uses the test-weighted mean of tile download speeds. The solid series gives equal weight to the same 206 municipalities; the dashed grey series uses all 210–213 municipalities available in each quarter. The shaded region is the cross-municipality P25–P75 range in the fixed sample. Quarterly averages are plotted at quarter midpoints. Vertically labeled lines mark December 1, 2021 provisional assignments, March 17, 2022 auction completion, July 1, 2022 licence validity, and July 31, 2023 release of the 700/800 MHz bands from analogue TV use. The auction interval, March 8–17, is shaded. The endpoints are 2021Q1 and 2024Q4.

4.2 Initially slow municipalities: improvement and remaining gaps

Figure 2 follows groups fixed by 2019 performance. The initially slowest group's geometric mean of tile-median speeds rises from 15.6 to 27.4 Mbps; the fastest group's rises from 24.3 to 37.4. The fastest-to-slowest ratio falls from 1.56 to 1.37, while the absolute difference rises from 8.8 to 10.1 Mbps. The comparison concerns the same statistic in both calculations.

The low-speed-share difference also narrows. The slowest–fastest gap below 10 Mbps falls from 12.9 to 5.9 percentage points; the corresponding below-25 gap falls from 14.0 to 7.8. Thus initially slow municipalities improve both in speed and in the prevalence of very low recorded performance. The faster group's larger Mbps increase widens the absolute speed difference over the same period.

Table 2 asks how changes relate to initial speed continuously, conditional on province effects. A one-standard-deviation higher initial speed is associated with 2.21 and 2.24 percentage points less reduction in the below-10 and below-25 shares. Wild-cluster probabilities

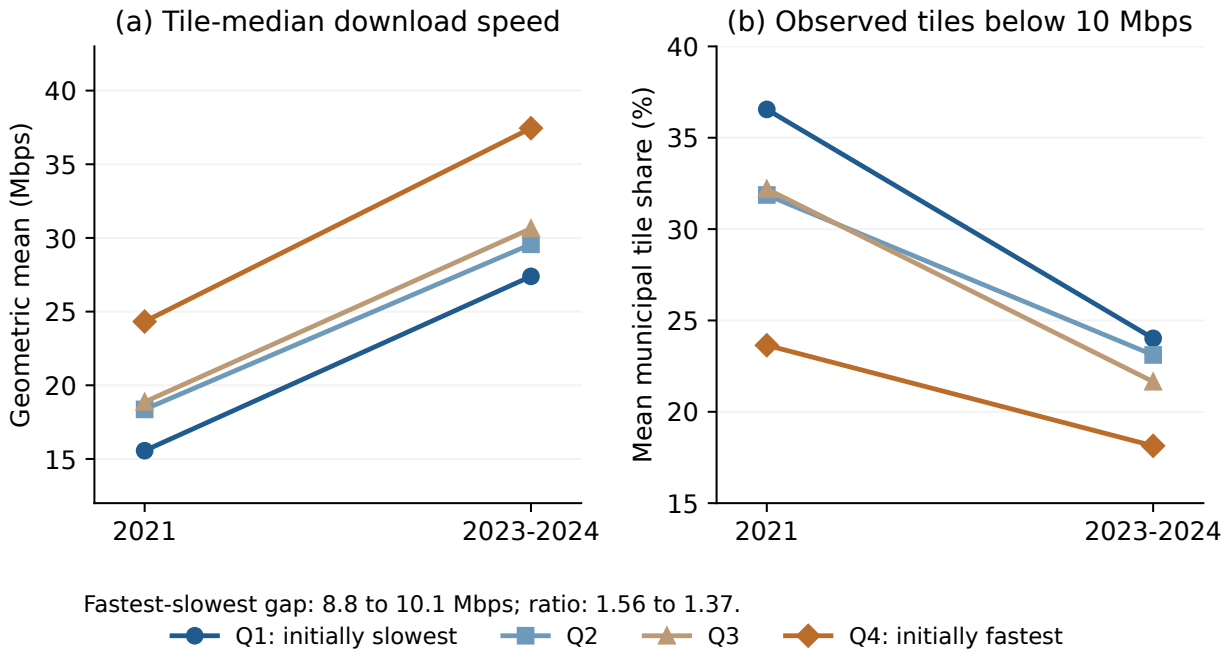


Figure 2. Performance changes by fixed 2019 initial-speed quartile

Notes: The same 206 municipalities are grouped by mean log municipal download speed over observed 2019 quarters; group sizes are 52/51/51/52. Points compare the four-quarter 2021 average with the eight-quarter 2023–2024 average, giving municipalities and quarters equal weight. Panel (a) exponentiates mean log municipality-quarter tile-median speed; it is not the arithmetic municipal mean in Figure 3. Panel (b) reports mean municipal shares of observed tiles below 10 Mbps. Annotation gaps and ratios use unrounded panel-(a) values. Lines join two period summaries and are not quarterly trajectories.

are 0.023 and 0.047, but both six-outcome adjusted values are 0.141. Mean-log-speed and high-speed-share coefficients are imprecise. The estimates suggest greater reductions in very low performance among initially slower places, with limited statistical support after the six-outcome adjustment. Requiring all four 2019 quarters preserves this direction and the uncertainty after adjustment (Appendix Table A2).

Table 2. Initial performance and subsequent changes

	Download speed			Observed tile shares		
Outcome	Mean (log) (1)	P25 (log) (2)	Median (log) (3)	< 10 (pp) (4)	< 25 (pp) (5)	≥ 50 (pp) (6)
Initial speed (1 SD)	-0.0105	-0.0298	-0.0508	2.2095**	2.2441**	0.2300
	(0.0240)	(0.0325)	(0.0327)	(0.7348)	(0.8448)	(0.7875)
Cluster p	0.672	0.385	0.159	0.017	0.029	0.778
Wild-cluster p	0.656	0.406	0.160	0.023	0.047	0.859
BH-adjusted q	0.787	0.609	0.320	0.141	0.141	0.859
Province FE	Yes	Yes	Yes	Yes	Yes	Yes
Unit of analysis	Municipality	Municipality	Municipality	Municipality	Municipality	Municipality
Observations	206	206	206	206	206	206
Municipalities	206	206	206	206	206	206
Province clusters	9	9	9	9	9	9

Notes: Equation (3), 206 municipalities, 2021 versus 2023–2024. The regressor is a one-standard-deviation increase in mean log download speed over observed 2019 quarters. Province fixed effects are included. SE is the province-cluster CR1 standard error; cluster p uses a t distribution with eight degrees of freedom. Small-sample p is the restricted wild-cluster bootstrap result. BH q adjusts that result across all six outcomes. Stars indicate $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$, before BH adjustment. All share coefficients are percentage points.

4.3 What changed in regulatory priority areas?

Figure 3 compares the same 92 Batch 3 and 114 other municipalities in 2021 and 2023–2024. Mean speed rises by 17.14 Mbps in Batch 3, compared with 27.20 elsewhere. The priority group's average-speed shortfall therefore increases from 6.57 to 16.64 Mbps.

The lower tail gives a different comparison. Batch 3's below-10-Mbps share falls by 11.00 percentage points, compared with 7.99 elsewhere. Its disadvantage on this measure narrows from 7.02 to 4.01 percentage points. Yet the at-least-50-Mbps share grows by 14.58 points in Batch 3 and 18.38 elsewhere, widening its shortfall from 3.31 to 7.12 points. The priority group experiences a larger reduction in very low performance alongside smaller gains in average speed and high-speed expansion.

Within the same priority group, the reduction in very slow observations exceeds that elsewhere, alongside smaller gains in average speed and high-speed shares. These measures describe the service patterns observed in priority areas; ICASA's cell-edge and coverage obligations use separate measures. The supplementary Early–Late DDD in Appendix B compares different spectrum-access histories.

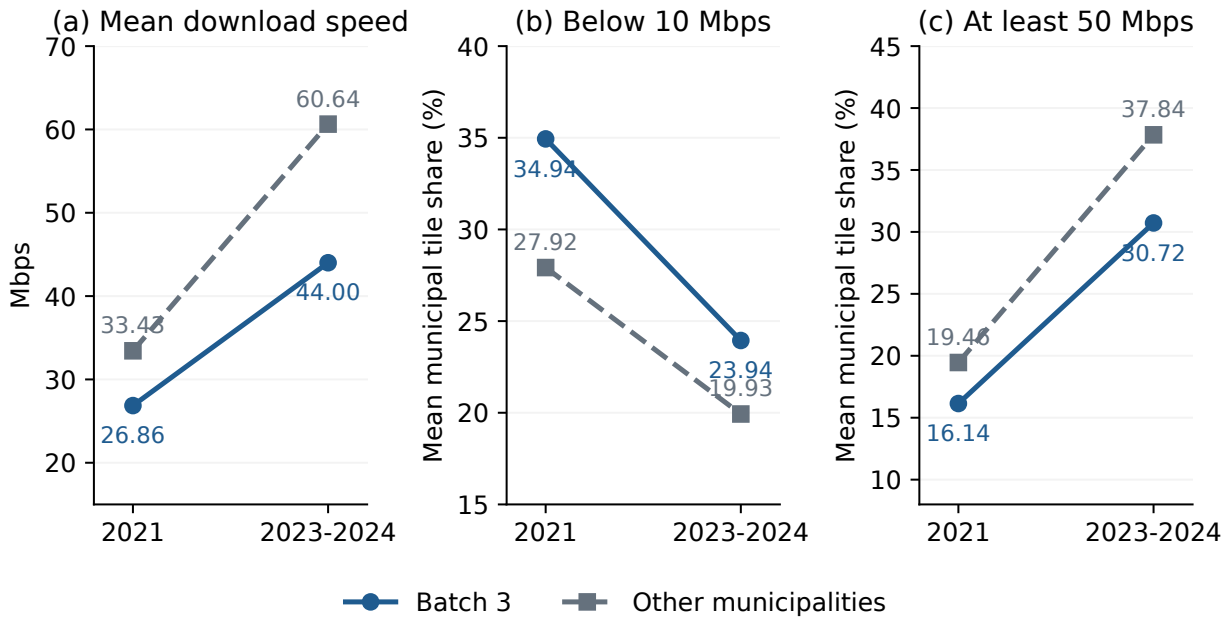


Figure 3. Regulatory priority and three aspects of upgrading

Notes: Descriptive period averages for the unchanged 206-municipality sample: Batch 3 $N = 92$ and other municipalities $N = 114$. The periods are 2021 and 2023–2024; quarters and municipalities receive equal weight. Panel (a) averages test-weighted municipal mean download speeds arithmetically. Panels (b)–(c) average municipal tile shares, not resident shares. Exact labels are rounded to two decimals. The lines join period summaries. There are no estimated confidence intervals or tests of designation effects. Appendix Table A3 retains all six original outcomes and changes.

5 Poverty and Broadband Upgrading

5.1 The socioeconomic gradient

Initial service quality and economic disadvantage need not rank municipalities in the same way. We therefore use the separate 223-municipality poverty frame to ask how improvements differ with baseline poverty. Figure 4 removes province means from both axes. Small points are municipalities, larger points summarize ten bins, and the fitted line corresponds to Equation (4). The slope summarizes the within-province association between baseline poverty and subsequent changes.

Ten percentage points more baseline poverty is associated with 3.48 percentage points less expansion of the at-least-50-Mbps share (Table 3). The CR1 interval is $[-7.43, 0.47]$ and its p-value is 0.077. The restricted wild-cluster value is 0.023; BH adjustment across all six outcomes gives 0.105. The coefficient remains exploratory. The earlier two-threshold family gives 0.047, but its selection before results was not documented.

The below-10-Mbps coefficient is -1.92 percentage points, with wild $p = 0.449$. The mean-log-speed coefficient is -0.0980, with wild $p = 0.035$ and six-outcome $q = 0.105$.

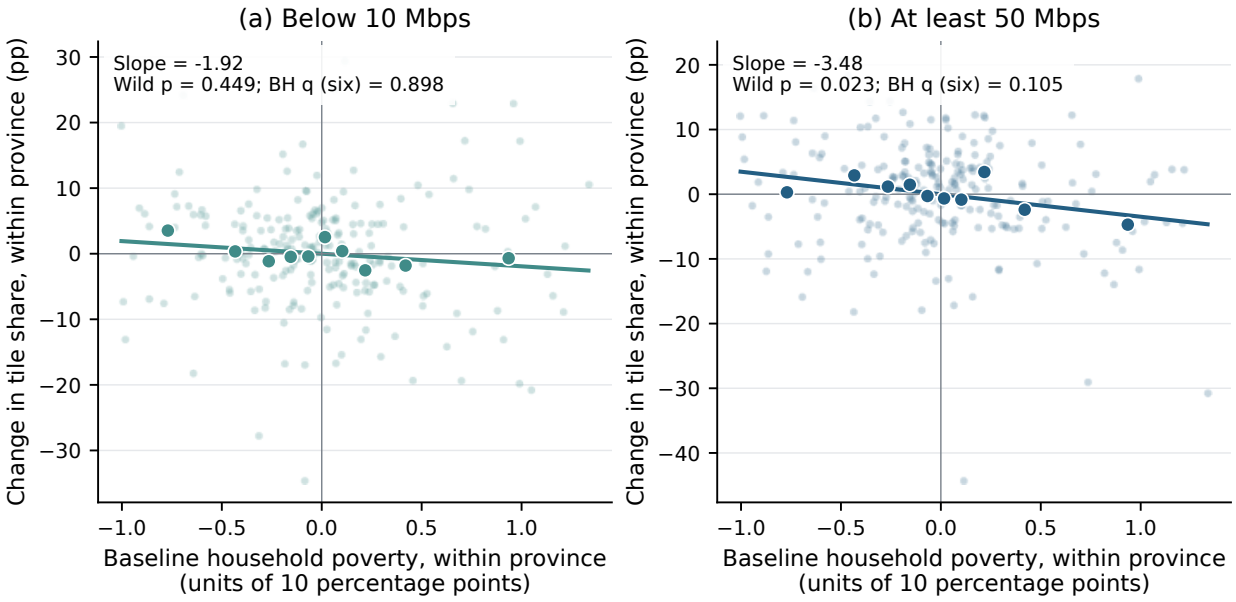


Figure 4. Baseline poverty and within-province performance changes

Notes: The underlying sample contains 223 municipalities on 2011 boundaries. The horizontal variable is $H_{2016}/10$ after subtracting its province mean. Vertical variables are changes in observed tile shares in percentage points, 2023–2024 minus 2021, also demeaned within province. Small points represent municipalities; large points are means within ten bins of the horizontal variable. Lines represent the continuous regression coefficients. No confidence band is plotted. The annotations report wild-cluster p-values and BH q-values across all six poverty outcomes. Table 3 also retains the earlier family grouping.

Table 3 reports the full set of poverty gradients. The low-speed gap reduction in Figure 2 concerns groups defined by initial speed, providing a separate comparison from baseline poverty.

Table 3. Baseline poverty and subsequent performance changes

	Download speed			Observed tile shares		
Outcome	Mean (log) (1)	P25 (log) (2)	Median (log) (3)	< 10 (pp) (4)	< 25 (pp) (5)	≥ 50 (pp) (6)
Baseline poverty (10 pp)	-0.0980**	0.0049	-0.0266	-1.9212	0.4651	-3.4798**
	(0.0473)	(0.0789)	(0.0517)	(1.4899)	(1.0559)	(1.7128)
Cluster p	0.072	0.952	0.621	0.233	0.671	0.077
Wild-cluster p	0.035	0.914	0.621	0.449	0.840	0.023
BH q (all six)	0.105	0.914	0.914	0.898	0.914	0.105
BH q (earlier 2+4)	0.141	0.914	0.914	0.449	0.914	0.047
Province FE	Yes	Yes	Yes	Yes	Yes	Yes
Unit of analysis	Municipality	Municipality	Municipality	Municipality	Municipality	Municipality
Observations	223	223	223	223	223	223
Municipalities	223	223	223	223	223	223
Province clusters	9	9	9	9	9	9

Notes: Equation (4), 223 municipalities on 2011 boundaries. The regressor is ten percentage points of the 2016 poverty headcount. Changes compare 2021 with 2023–2024, conditional on province fixed effects. SE and cluster p use province CR1 inference and eight degrees of freedom; small-sample p uses the wild-cluster result. BH values are shown across all six outcomes and, for comparison, under the earlier two-threshold/four-other grouping. Pre-results family selection is not documented. These within-family adjustments do not cover the other analytical choices in the paper. Stars use unadjusted wild p as defined in Table 2.

5.2 Conditioning on initial service quality

Adding mean log download speed in 2019 leaves the estimated poverty gradients similar in direction and magnitude. In the same 223-municipality sample with province effects, the high-speed-share coefficient changes from -3.4798 to -3.2291 percentage points per ten-percentage-point increase in baseline poverty. The mean-log-speed coefficient changes from -0.0980 to -0.0952. Appendix Table A11 compares all six outcomes under the original and augmented specifications.

Statistical support varies across inference procedures. For high-speed share and mean log speed, respectively, wild-cluster p-values are 0.0117 and 0.0313, and six-outcome BH q-values are 0.0703 and 0.0938. Both meet the ten-percent BH threshold, with CR1 p-values of 0.0546 and 0.0550 and 95 percent intervals that include zero. The other four coefficients remain imprecise under both CR1 and wild-cluster inference. The check therefore supports a similar estimated pattern after conditioning on initial mean speed; the five-percent evidence remains sensitive to inference and multiplicity adjustment.

5.3 Sensitivity and which places can be observed

The high-speed coefficient stays negative when Limpopo is excluded, when available observations are used, and when 2021 is compared with 2024 alone. Its precision changes (Appendix Table A9). A poverty-quartile specification gives an imprecise most-poor versus least-poor

contrast, and the middle groups do not form a smooth monotonic sequence (Appendix Figure A5 and Table A10). Negative dynamic high-speed gradients also appear before the 2022Q3 licence marker, with wide pointwise intervals (Appendix Figure A4). The negative pre-marker gradients place the subsequent poverty association in a longer pattern of differing changes.

Observation support is uneven across poverty groups. Figure 5 reports the share of municipalities with at least one tile measured in every quarter. The share is 69.6 percent in the least-poor quartile and 19.6 percent in the poorest. Restricting attention to fixed locations therefore changes which places can enter the comparison.

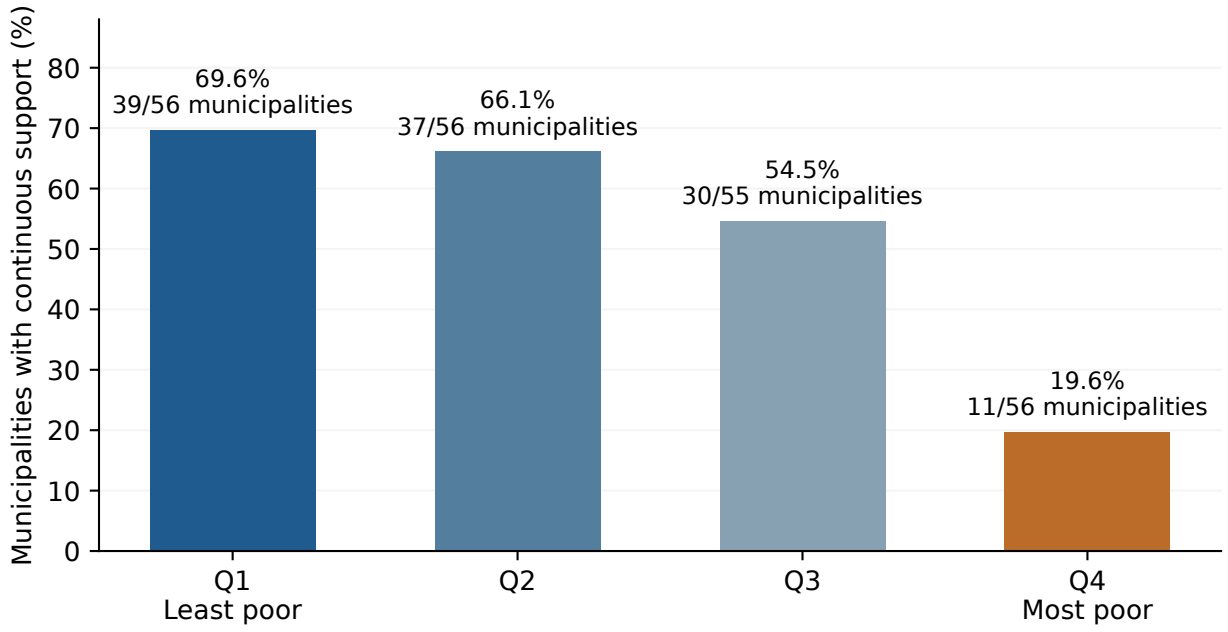


Figure 5. Continuous-location support by baseline-poverty quartile

Notes: The denominator is the unchanged 223-municipality poverty sample, grouped by fixed 2016 household-poverty quartiles of 56/56/55/56 municipalities. A supported municipality has at least one tile observed in all sixteen quarters, 2021Q1–2024Q4. Counts are 39/37/30/11, totaling 117 municipalities. Bars are the corresponding municipal support rates, not household observation rates or internet coverage. Source: the existing measurement audit in Appendix Table A12.

In the 117-municipality supported subset, the high-speed poverty coefficient is -7.75, with a CR1 interval of $[-29.23, 13.73]$ and wild $p = 0.516$. The wide interval and selected coverage leave the main gradient unresolved in this subset. Regressions of changes in tile count, test count, and tests per tile are also imprecise (Appendix Table A12). Fixed locations hold geography constant but leave testers, devices, subscriptions, and test-taking choices free to change.

6 Economic Interpretation and Policy Implications

ICASA-designated priority municipalities experienced a larger decline in the share of observed tiles below 10 Mbps than other municipalities. Other municipalities gained more in average speed and in the share of tiles reaching at least 50 Mbps. These changes answer two separate policy questions: whether weak service improved in priority areas, and whether those areas narrowed their performance gaps with other places. The first comparison records progress. The second shows wider gaps in average speed and high-speed performance. The same priority group thus experienced an improvement in weak service alongside growing shortfalls on other measures.

The economic significance of these changes depends on how users value better service at different starting speeds. Ahlfeldt, Koutroumpis, and Valletti (2017) find diminishing returns to speed in household valuation, and Fabling and Grimes (2021) find clearer productivity gains where firms make complementary investments. These findings give a reason to examine the location and type of service improvement together. They also show why the welfare ranking depends on which activities existing connections constrain and how users respond. Our speed measures provide evidence on service conditions; measuring productivity or welfare gains would require corresponding economic outcomes and evidence on use.

Operators may prioritize locations where expected subscription revenues are high relative to deployment costs. On the user side, affordability, skills, devices, and organizational changes may affect demand for better service and its productive use. These incentives offer possible explanations for the geography of upgrading. Distinguishing their roles would require local investment, subscription, and device data. The broader evidence on infrastructure location and economic gains (Duflo and Pande, 2007) reinforces the relevance of connecting service improvements to these local conditions.

The poverty analysis adds a socioeconomic perspective. Higher baseline poverty is associated with less high-speed expansion, and conditioning on initial mean speed leaves the estimated gradient similar in size. Its statistical support varies across inference procedures. Measurement support is also uneven: continuously observed locations are especially sparse in poorer municipalities. The results therefore describe selected tests and locations, with population representativeness unresolved. Holding locations fixed still allows the composition of testers, devices, and subscriptions to change.

For policy monitoring, weak-service incidence, average performance, and high-speed shares can be reported together for the same geographic groups. Initial quality, poverty, and regulatory priority each define a different comparison. Following these groups over time would reveal whether improvements within underserved areas are accompanied by narrower gaps with other places. The Batch 3 evidence makes this distinction concrete: reducing very slow service and narrowing higher-speed gaps can yield different assessments of the same

upgrading period. Choosing among rollout, affordability, or device interventions would additionally require evidence on their costs and use.

7 Conclusion

South Africa's mobile broadband improvements changed different service disadvantages in different ways. Initially slow municipalities reduced their low-speed-share gap and relative speed ratio, alongside a wider absolute Mbps gap. Regulatory priority areas experienced a larger reduction in very slow observations than other municipalities, yet their shortfalls in average speed and high-speed shares grew. The separate, exploratory poverty comparison associates higher baseline poverty with less high-speed expansion. Adding 2019 initial mean speed leaves that estimate similar in size.

Following several performance measures for the same places reveals how progress in weak service can coexist with widening gaps in higher-speed performance. The regulatory priority comparison connects this distinction to an existing policy classification. Broadband evaluation can use it to assess both improvement within underserved places and changes in their position relative to other places, with the measurement coverage of each comparison made explicit.

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A Data Construction and Coverage

The 2018-boundary performance frame and the 2011-boundary poverty frame aggregate the same underlying quarterly tile measurements separately. Tile centroids locate observations in each municipal geography. The 2011 geometry is distributed through Africa Open Data with MDB municipal codes. Polygon repair is used where needed. Drawing simplification does not change the analytical assignment. Because municipal boundaries changed, the two frames differ in both their number of units and their spatial aggregation.

Table A1. Missing quarters in the 2018 municipal frame

Code	Municipality	Province	Missing quarters
EC135	Intsika Yethu	EC	2021Q3,2024Q1
EC137	Engcobo	EC	2021Q1
EC138	Sakhisizwe	EC	2024Q1
EC444	Ntabankulu	EC	2021Q3
KZN286	Nkandla	KZN	2021Q1
KZN293	Ndwedwe	KZN	2023Q2
NC074	Kareeberg	NC	2021Q1

Notes: The full frame contains 213 municipalities and sixteen quarters. These seven municipalities account for all eight missing municipality-quarter download observations in the current window. They remain in the all-available series whenever observed. The principal complete panel contains the other 206 municipalities.

The observed-footprint control retains its existing denominator: the set of tiles observed at least once in the earlier seventeen-quarter construction window. Its numerator is the currently observed tile count.

Limpopo's 25 headcounts follow the official 2016 correction reproduced in the indexed revised municipal table. We rechecked the 25 entries against the official PDF's accessible text in this revision. Direct retrieval of the revised PDF returned an access error, so an independent visual and binary-file check remains unresolved. The no-Limpopo estimates in Table A9 retain sensitivity to this source limitation. The official terminology defines the poverty headcount as the share of multidimensionally poor households.

Initial-performance regressions adjust all six outcomes together. The poverty table now shows both six-outcome BH values and the earlier two-threshold/four-other grouping. The earlier families were not documented as pre-analysis choices. Poverty-quartile sensitivity tables retain their earlier contrast families and are labeled accordingly. Policy Tables A4 and A6 report wild-cluster probabilities and six-outcome BH values within each model. The earlier time-window and classification tables retain their stated conventions. None is a paper-wide multiplicity correction. Reallocation exercises remain descriptive sensitivity diagnostics.

A.1 Initial-performance baseline information

Table A2 restricts the initial-performance comparison to the 198 municipalities with all four 2019 quarters observed. It addresses an unequal-baseline-information concern while preserving the 2021 versus 2023–2024 outcome comparison. The below-10-Mbps direction persists, but the adjusted probability remains 0.141.

Table A2. Initial-speed association with four observed baseline quarters

	Download speed			Observed tile shares		
Outcome	Mean (log)	P25 (log)	Median (log)	< 10 (pp)	< 25 (pp)	≥ 50 (pp)
	(1)	(2)	(3)	(4)	(5)	(6)
Initial speed (1 SD)	-0.0069	-0.0457	-0.0651	2.8274**	2.4160*	0.5970
	(0.0371)	(0.0377)	(0.0335)	(0.9574)	(0.9800)	(1.1245)
Cluster p	0.857	0.260	0.088	0.018	0.039	0.610
Wild-cluster p	0.859	0.285	0.121	0.023	0.066	0.758
BH-adjusted q	0.859	0.428	0.242	0.141	0.199	0.859
Province FE	Yes	Yes	Yes	Yes	Yes	Yes
Unit of analysis	Municipality	Municipality	Municipality	Municipality	Municipality	Municipality
Observations	198	198	198	198	198	198
Municipalities	198	198	198	198	198	198
Province clusters	9	9	9	9	9	9

Notes: 198 municipalities and nine provinces. Same province-effects change regression and units as Table 2; the sole sample change requires all four 2019 baseline quarters. Small-sample probabilities are wild-cluster results; BH covers all six outcomes. The baseline variable retains its full-main-sample standardization.

B Institutional Comparisons and Sensitivity

Appendix Figure A1 locates the five Early provinces and the regulatory priority municipalities. Early is Free State, Limpopo, Mpumalanga, North West, and Northern Cape; Late is Eastern Cape, Gauteng, KwaZulu-Natal, and Western Cape. A July 2022 government statement identifies the earlier-clearance group.[7] Early provinces contain 41 Batch 3 and 60 other sample municipalities; Late provinces contain 51 Batch 3 and 54 others.

Table A3 retains the six descriptive outcomes underlying main Figure 3. It is distinct from the Early–Late comparison below.

[7]Department of Communications and Digital Technologies, July 7, 2022.

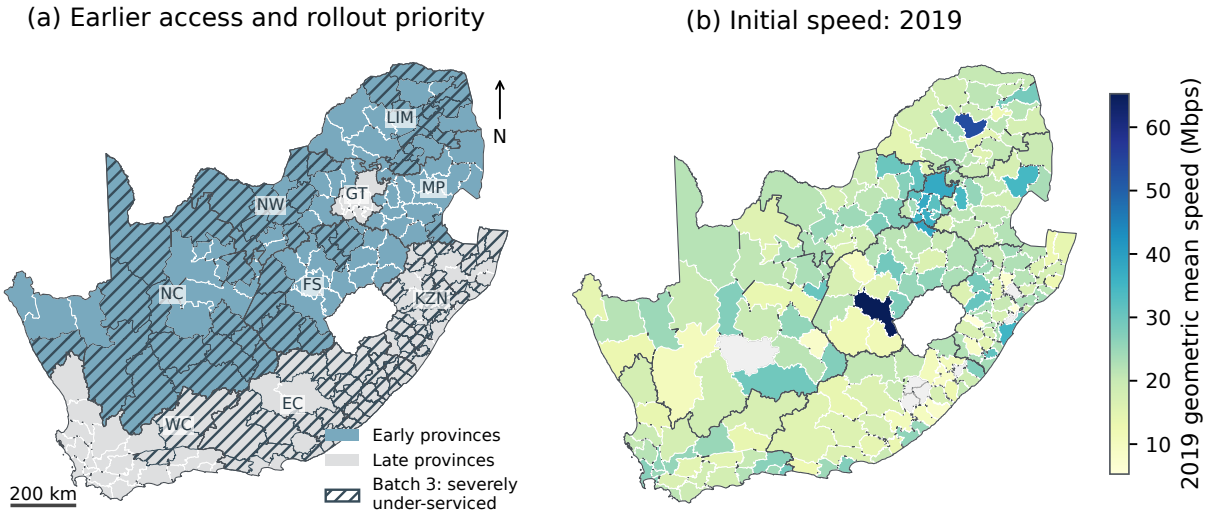


Figure A1. Policy geography and initial mobile performance

Notes: Panel (a) uses all 213 municipalities on the 2018 Municipal Demarcation Board geography. Blue identifies Early provinces; hatching identifies the 99 severely under-serviced Batch 3 municipalities. Panel (b) shows the geometric mean of quarterly municipal download speeds over observed 2019 quarters for the 206 municipalities in the main sample; grey denotes the seven excluded municipalities. Mean speeds within a quarter are test-weighted. Maps use an Albers equal-area projection centered on South Africa; boundaries are simplified only for drawing. Sources: local MDB geometry, ICASA municipal schedule, and Ookla analytical panel.

Table A3. Regulatory priority and observed performance changes

	Batch 3 ($N = 92$)			Other ($N = 114$)		
Outcome	2021	2023–24	Change	2021	2023–24	Change
Mean speed (Mbps)	26.86	44.00	+17.14	33.43	60.64	+27.20
Tile median (Mbps)	19.23	30.73	+11.51	22.37	36.85	+14.48
Tile P25 (Mbps)	9.57	14.53	+4.97	10.02	15.20	+5.18
Below 10 Mbps (%)	34.94	23.94	-11.00	27.92	19.93	-7.99
Below 25 Mbps (%)	62.31	46.03	-16.28	55.26	40.38	-14.88
At least 50 Mbps (%)	16.14	30.72	+14.58	19.46	37.84	+18.38

Notes: The same 206 municipalities enter both periods: 92 Batch 3 and 114 other municipalities. Pre is the four-quarter 2021 average; post is the eight-quarter 2023–2024 average. Municipality-quarter means retain the original within-quarter test weighting. Period and group means give equal weight to quarters and municipalities. Tile P25 and median rows average the municipal-quarter quantiles arithmetically. Share levels are percentages; their changes are percentage points. These descriptive differences have no inferential stars and are not estimates of designation effects or licence compliance. Source: the unchanged policy panel and ICASA municipal classification.

The preceding analyses characterize the distribution of observed upgrading by initial service quality and poverty. We next ask a separate institutional question: does the Batch 3–other-municipality improvement gap differ between Early and Late provinces? This supplementary triple-difference comparison relates performance paths to earlier spectrum access and regulatory priority. Its classifications differ from the preceding baselines. The exercise therefore neither identifies nor explains the initial-speed and poverty gradients.

B.1 The triple-difference estimand

The policy comparison asks whether the Batch 3–other-municipality improvement gap is larger in Early than in Late provinces. It conditions on observed tile count and footprint. The coefficient summarizes relative changes in performance under these measurement conditions. In the principal scalar model, $Post_t$ equals one in 2023–2024 and zero in 2021. Observations from 2022 are excluded. We estimate

$$Y_{ipt} = \alpha_i + \lambda_{pt} + \mu_{B_i,t} + \beta(Early_p \times B_i \times Post_t) + X'_{it}\gamma + \varepsilon_{ipt}. \tag{5}$$

Here B_i is the Batch 3 indicator. Municipality effects absorb fixed differences across places. Province-by-quarter effects λ_{pt} absorb common changes within a province. Batch-3-status-by-quarter effects $\mu_{B_i,t}$ allow nationwide priority-group differences to vary each quarter. The controls X_{it} contain log valid tile count and observed tile footprint share.

Measurement activity can respond to spectrum availability and service improvements. Conditioning on policy-affected variables can remove part of a total effect or introduce selection bias (Cinelli, Forney, and Pearl, 2024). We retain the basic controlled specification as M1 and remove both measurement controls in M2. M3 adds an Early × Batch 3 linear differential trend to M1; M4 adds the same trend to M2. All four models use the same 206 municipalities, twelve quarters, treatment assignments, and pre/post dates. This grid checks control and trajectory sensitivity without selecting a new baseline equation.

A causal interpretation would additionally require a stable Early–Late difference in Batch 3 relative trends (Olden and Møen, 2022). This condition concerns how those trends would evolve in the absence of the earlier-access history. Province-by-quarter effects absorb province-wide changes. The remaining variation comes from Batch 3 relative to other municipalities within each province. Differential deployment, broadcasting relocation, and changes in measurement composition could affect these relative trends.

The event-study version replaces the post interaction with separate Early × Batch 3 interactions for each quarter. It retains all sixteen quarters and normalizes to 2021Q4. The scalar comparison uses the full-year 2021 average as its baseline.

B.2 Inference and results

Table A4 reports the unchanged basic specification with CR1, restricted wild-cluster, and six-outcome BH probabilities. There are nine province clusters, so even bootstrap inference has limited support. Appendix Table A6 applies the same reporting convention to all four specifications. The adjustments are within each six-outcome model; they do not cover the four-model search. Appendix B retains the separate province-classification diagnostic over all 126 five-province groupings.

The basic mean and median coefficients remain 0.200 and 0.117 log points. Their wild-cluster p-values are 0.043 and 0.051, and both six-outcome BH values are 0.152 (Table A4). Removing the measurement controls gives 0.202 and 0.122. Thus the positive central-performance contrast does not depend strongly on including these two controls in the model without a differential trend. This check does not establish that the controls are exogenous or that the contrast is causal.

Table A4. Conditional performance changes by earlier access and regulatory priority

	Download speed			Observed tile shares		
Outcome	Mean (log)	P25 (log)	Median (log)	< 10 (pp)	< 25 (pp)	≥ 50 (pp)
	(1)	(2)	(3)	(4)	(5)	(6)
Early × Batch 3 × Post	0.1998	-0.0085	0.1171	1.0169	-0.0639	3.0379
	(0.0653)	(0.0864)	(0.0375)	(1.2707)	(2.5310)	(1.7674)
CR1 p (unadjusted)	0.016	0.924	0.014	0.447	0.980	0.124
Wild-cluster p	0.043	0.938	0.051	0.488	0.984	0.199
BH q (six outcomes)	0.152	0.984	0.152	0.732	0.984	0.398
Municipality FE	Yes	Yes	Yes	Yes	Yes	Yes
Province × Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes
Batch 3 status × Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes
Log valid tile count	Yes	Yes	Yes	Yes	Yes	Yes
Observed tile footprint share	Yes	Yes	Yes	Yes	Yes	Yes
Differential linear trend	No	No	No	No	No	No
Observations	2,472	2,472	2,472	2,472	2,472	2,472
Municipalities	206	206	206	206	206	206
Province clusters	9	9	9	9	9	9

Notes: Equation (5); 206 municipalities, 12 quarters, and 2,472 observations. Pre is 2021 and post is 2023–2024; 2022 is excluded. Fixed effects and contemporaneous measurement controls are listed below the estimates. Parentheses contain province CR1 standard errors. CR1 p-values use $t(8)$. Restricted wild-cluster p-values enumerate 512 province sign patterns; BH q-values adjust them across these six outcomes. These are within-model adjustments. Table A7 reports the separate province-classification diagnostic.

Appendix Figure A2 shows the quarterly institutional contrasts. The mean and median paths become positive around the transition, but are uneven. Its intervals are pointwise CR1 intervals. Non-rejection in the short pre-period does not establish parallel trends (Roth, 2022).

Differential trends matter more for magnitude. The mean coefficient is 0.468 with controls and 0.440 without them. In the four models, its wild-cluster p-values range from 0.016 to 0.047 and its six-outcome BH values from 0.094 to 0.164. None meets a five-percent BH threshold. The positive sign is common to this grid, but the size of the contrast depends on the assumed trajectory. The all-sixteen-quarter comparison without the additional trend yields 0.188. Appendix B retains those window checks and reports the full new control–trend grid.

A separate broadcasting-migration diagnostic uses 63 municipalities in the four Late provinces. It interacts above-694-MHz exposure with the period from 2023Q3. Its mean-speed coefficient is -0.125. This within-province exposure comparison describes a different margin of the final transition. It does not reproduce the earlier-access comparison's positive central-performance pattern (Appendix Table A8).

B.3 Dynamic contrasts and additional checks

Table A5 changes the treatment of time in the policy comparison. The first panel adds the differential linear trend to the principal twelve-quarter sample. The second includes all sixteen quarters with post beginning in 2022Q3. The third also adds the differential trend to that sixteen-quarter model. Appendix B discusses the larger mean estimate in the first panel.

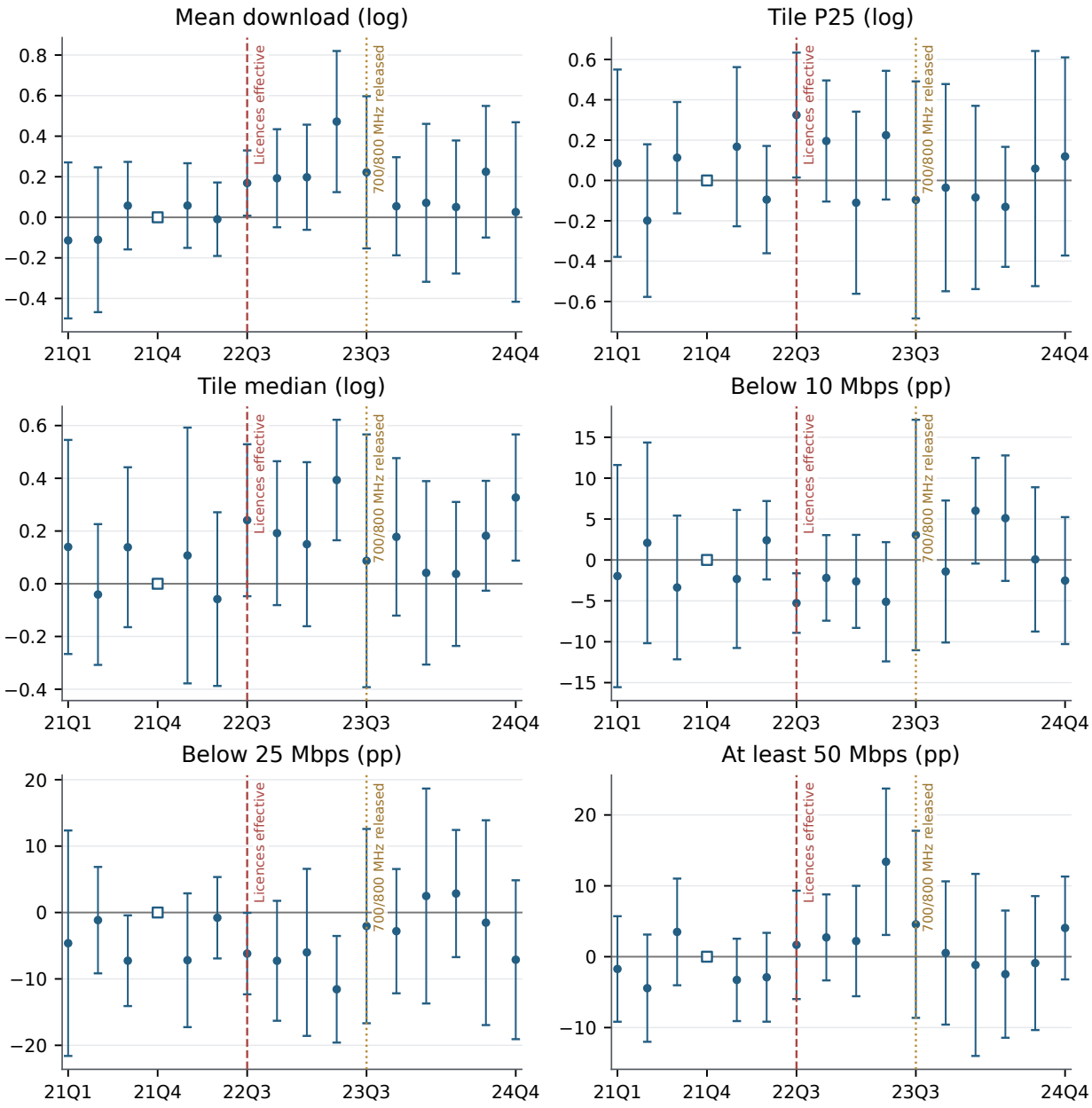


Figure A2. Dynamic Early × Batch 3 contrasts

Notes: Event-study estimates using all 16 quarters and 206 municipalities. The omitted quarter is 2021Q4, shown by the open square at zero; it has no estimated interval. Event time zero is 2022Q3. The specification includes the fixed effects and observation controls in Table A4. Error bars are pointwise 95 percent CR1 intervals using the $t(8)$ critical value. The red dashed line marks licence validity; the gold dotted line marks the 2023Q3 release of the 700/800 MHz bands. Share coefficients are percentage points.

Table A5. Conditional DDD comparisons across transition windows and trends

	Download speed			Observed tile shares		
Outcome	Mean (log)	P25 (log)	Median (log)	< 10 (pp)	< 25 (pp)	≥ 50 (pp)
	(1)	(2)	(3)	(4)	(5)	(6)
A. 2021 vs. 2023–2024; differential trend						
Early × Batch 3 × Post	0.4675	-0.0700	0.1608	-3.6259	-7.8720	10.2405
	(0.0929)	(0.3354)	(0.1774)	(5.5184)	(5.8536)	(4.2624)
CR1 p (unadjusted)	0.001	0.840	0.391	0.530	0.216	0.043
Observations	2,472	2,472	2,472	2,472	2,472	2,472
Province clusters	9	9	9	9	9	9
Differential linear trend	Yes	Yes	Yes	Yes	Yes	Yes
B. All 16 quarters; post from 2022Q3						
Early × Batch 3 × Post	0.1875	0.0349	0.1353	0.0351	-0.4227	3.9418
	(0.0549)	(0.0868)	(0.0484)	(1.4124)	(2.2484)	(1.6920)
CR1 p (unadjusted)	0.009	0.699	0.023	0.981	0.856	0.048
Observations	3,296	3,296	3,296	3,296	3,296	3,296
Province clusters	9	9	9	9	9	9
Differential linear trend	No	No	No	No	No	No
C. All 16 quarters; differential trend						
Early × Batch 3 × Post	0.2752	0.1823	0.1955	-4.7654	-5.4493	7.1593
	(0.0675)	(0.1855)	(0.0714)	(3.1175)	(2.1900)	(2.1279)
CR1 p (unadjusted)	0.004	0.355	0.026	0.165	0.038	0.010
Observations	3,296	3,296	3,296	3,296	3,296	3,296
Province clusters	9	9	9	9	9	9
Differential linear trend	Yes	Yes	Yes	Yes	Yes	Yes
Municipality FE	Yes	Yes	Yes	Yes	Yes	Yes
Province × Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes
Batch 3 status × Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes
Log valid tile count	Yes	Yes	Yes	Yes	Yes	Yes
Observed tile footprint share	Yes	Yes	Yes	Yes	Yes	Yes

Notes: 206 municipalities throughout. Panel A uses 2021 versus 2023–2024; B and C use all sixteen quarters and post from 2022Q3. Trend models add an Early × Batch 3 linear differential trend. Other fixed effects and contemporaneous measurement controls match the main model. Parentheses contain province CR1 standard errors; p-values use $t(8)$ and are unadjusted for multiple testing.

Table A6. Control and differential-trend sensitivity in the unchanged policy sample

	Download speed			Observed tile shares		
Outcome	Mean (log) (1)	P25 (log) (2)	Median (log) (3)	< 10 (pp) (4)	< 25 (pp) (5)	≥ 50 (pp) (6)
M1: Controls; no differential trend						
Early × Batch 3 × Post	0.1998	-0.0085	0.1171	1.0169	-0.0639	3.0379
	(0.0653)	(0.0864)	(0.0375)	(1.2707)	(2.5310)	(1.7674)
CR1 p	0.016	0.924	0.014	0.447	0.980	0.124
Wild p	0.043	0.938	0.051	0.488	0.984	0.199
BH q (six)	0.152	0.984	0.152	0.732	0.984	0.398
M2: No controls; no differential trend						
Early × Batch 3 × Post	0.2022	0.0333	0.1221	0.5324	-0.2215	3.5802
	(0.0666)	(0.0981)	(0.0393)	(1.2675)	(2.6214)	(1.9398)
CR1 p	0.016	0.743	0.015	0.686	0.935	0.102
Wild p	0.047	0.758	0.055	0.680	0.969	0.180
BH q (six)	0.164	0.909	0.164	0.909	0.969	0.359
M3: Controls; differential trend						
Early × Batch 3 × Post	0.4675	-0.0700	0.1608	-3.6259	-7.8720	10.2405
	(0.0929)	(0.3354)	(0.1774)	(5.5184)	(5.8536)	(4.2624)
CR1 p	0.001	0.840	0.391	0.530	0.216	0.043
Wild p	0.023	0.844	0.406	0.516	0.266	0.035
BH q (six)	0.105	0.844	0.609	0.619	0.531	0.105
M4: No controls; differential trend						
Early × Batch 3 × Post	0.4403	-0.0349	0.1496	-3.6428	-7.5187	10.6153
	(0.0940)	(0.3425)	(0.1750)	(5.4913)	(6.0769)	(4.4532)
CR1 p	0.002	0.921	0.417	0.526	0.251	0.044
Wild p	0.016	0.922	0.461	0.516	0.320	0.055
BH q (six)	0.094	0.922	0.619	0.619	0.619	0.164

Notes: Each panel uses the same 206 municipalities and 2,472 observations: 2021 versus 2023–2024, excluding 2022. Municipality, province-by-quarter, and Batch-3-status-by-quarter effects are included throughout. M1 and M3 include log valid tile count and observed footprint share; M2 and M4 remove both. M3 and M4 add the Early × Batch 3 linear trend. Parentheses are province CR1 standard errors. Wild probabilities use all 512 null-imposed Rademacher patterns; BH adjusts six outcomes separately within each panel, not the four specifications jointly.

B.4 Sensitivity to the classification of earlier-access provinces

Table A7 compares the historical Early grouping with all $\binom{9}{5} = 126$ ways of designating five provinces as Early. For each outcome and specification, we compare absolute coefficients across these combinations. The reallocation tail fraction is the share at least as large as the coefficient under the historical grouping. The historical assignment is included. The calculation compares absolute coefficients, with the outcome data and other specification choices held fixed. It is a descriptive measure of sensitivity to the grouping. The historical rollout does not establish that these 126 assignments were equally likely.

For mean download and the tile median in the principal specification, the fractions are 0.079 and 0.087. With a differential linear trend, the mean fraction is 0.008. This contrast reinforces the importance of the time specification when interpreting the position of the historical grouping among the alternatives.

Table A7. Sensitivity to the classification of earlier-access provinces

	Download speed			Observed tile shares		
Outcome	Mean (log) (1)	P25 (log) (2)	Median (log) (3)	< 10 (pp) (4)	< 25 (pp) (5)	≥ 50 (pp) (6)
2021 vs. 2023–2024	0.079	0.937	0.087	0.540	0.976	0.262
With differential trend	0.008	0.849	0.429	0.579	0.294	0.095
All 16 quarters	0.040	0.746	0.095	0.984	0.897	0.111
All 16 quarters + trend	0.040	0.421	0.087	0.262	0.063	0.032
Province classifications	126	126	126	126	126	126

Notes: Each cell is a reallocation tail fraction, defined in the text, over all 126 choices of five Early provinces. Rows use the same outcome definitions, fixed effects, controls, and time specifications as Tables A4 and A5. These fractions describe the placement of the historical coefficient among alternative classifications.

Figure A3 examines whether omitting one province materially changes three central coefficients. The error bars are the conventional cluster intervals using the remaining provinces, and the dashed line is the full-sample coefficient.

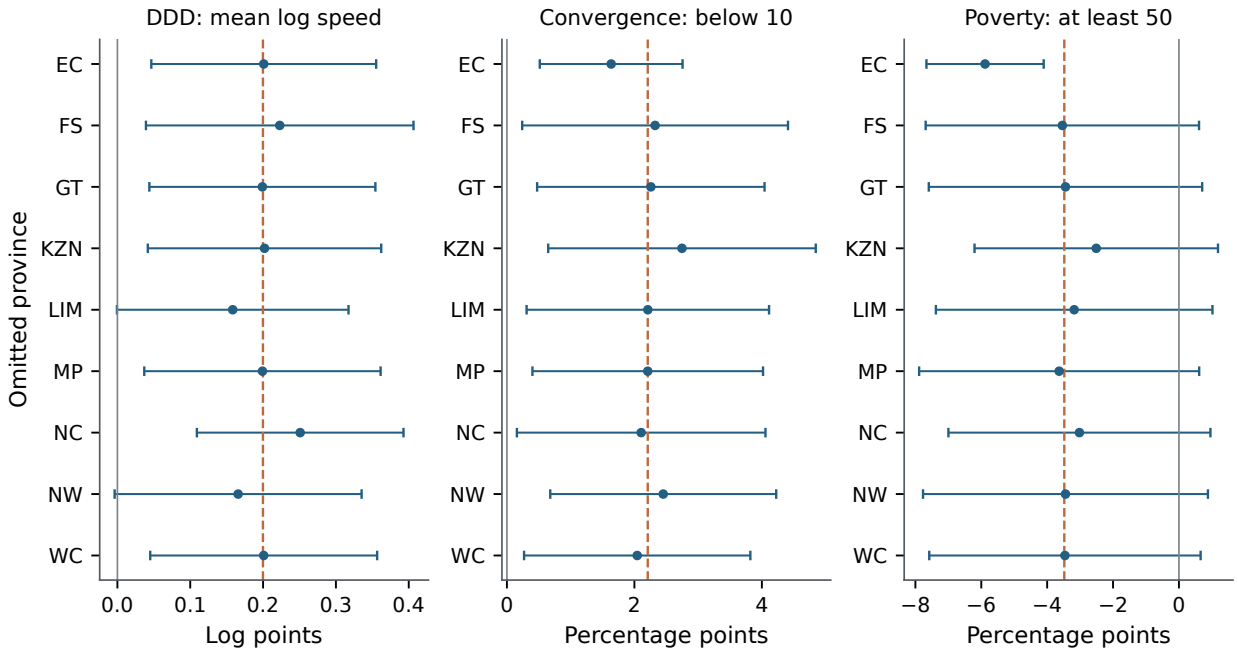


Figure A3. Influence of individual provinces on selected coefficients

Notes: Each row omits the named province. Panels show the principal DDD mean-log-speed coefficient, the initial-speed coefficient for the below-10 share, and the poverty coefficient for the at-least-50 share. Baseline definitions and pre/post windows are unchanged. Error bars are pointwise 95 percent CR1 intervals using the remaining province count. Orange dashes indicate the respective full-sample estimates. Units differ as labeled.

Table A8. Separate broadcasting-migration exposure diagnostic

	Download speed			Observed tile shares		
Outcome	Mean (log)	P25 (log)	Median (log)	< 10 (pp)	< 25 (pp)	≥ 50 (pp)
	(1)	(2)	(3)	(4)	(5)	(6)
Exposure × Post	-0.1250	-0.0885	-0.0533	0.9187	2.2683	-2.2766
	(0.0302)	(0.0798)	(0.0326)	(0.9823)	(2.5070)	(1.8597)
CR1 p (unadjusted)	0.026	0.348	0.201	0.419	0.432	0.308
Reallocation score	0.020	0.218	0.365	0.600	0.283	0.279
Municipality FE	Yes	Yes	Yes	Yes	Yes	Yes
Province × Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,008	1,008	1,008	1,008	1,008	1,008
Municipalities	63	63	63	63	63	63
Province clusters	4	4	4	4	4	4

Notes: 63 municipalities in four Late provinces. Above-694-MHz exposure is interacted with post from 2023Q3, with municipality and province-by-quarter effects. The first full quarter after July 31 clearance is 2023Q4. Parentheses contain province CR1 standard errors; p-values use $t(3)$ and are unadjusted for multiple testing. The reallocation score summarizes 1,999 within-province shuffles of municipal exposure: it is $(1 + k)/2{,}000$, where k counts shuffled coefficients at least as large in absolute value as the observed coefficient. It is reported as a descriptive sensitivity score.

C Poverty Gradients, Functional Form, and Sample Sensitivity

Figure A4 shows changes in the conditional poverty gradient quarter by quarter. The model contains municipality effects, province-by-quarter effects, and poverty-by-quarter interactions. Its normalization is 2021Q4. The principal scalar coefficient instead compares the average of all four 2021 quarters with the average of all eight 2023–2024 quarters. These estimates therefore answer different time-aggregation questions. The dynamic intervals use CR1 inference; the scalar table additionally reports the wild-cluster probabilities.

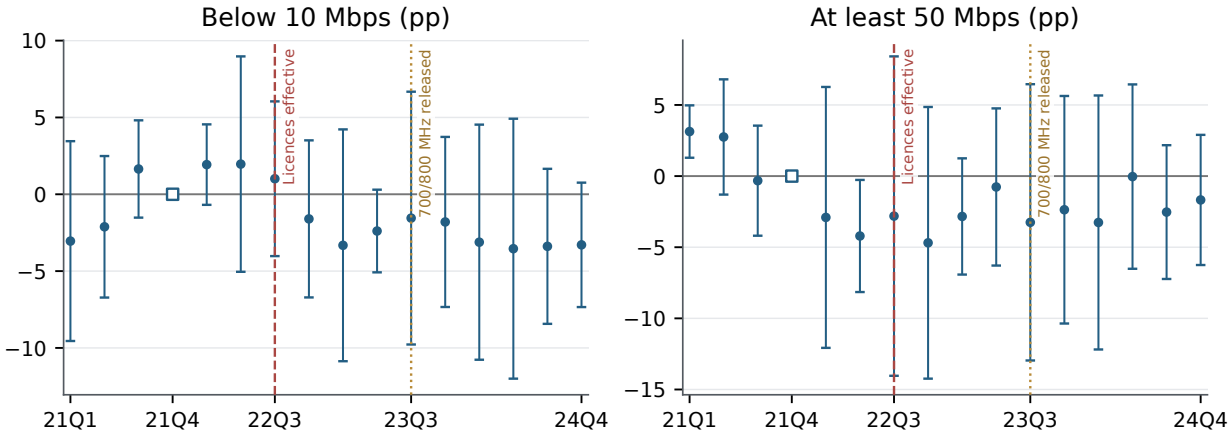


Figure A4. Dynamic conditional poverty gradients

Notes: 223 municipalities on 2011 boundaries, 2021Q1–2024Q4. Coefficients are on $H_{2016}/10$ interacted with each quarter, relative to 2021Q4, conditional on municipality and province-by-quarter effects. They describe percentage-point differences in changes associated with ten percentage points of baseline poverty. Open squares denote the omitted quarter. Bars are pointwise 95 percent CR1 intervals using $t(8)$.

Table A9 reports the full six-outcome results when excluding Limpopo, permitting available pre/post observations, and comparing 2021 with 2024 alone. These checks address source uncertainty, completeness, and post-period averaging, respectively. The high-speed coefficient stays negative, but adjusted values within the reported two-threshold family range from 0.047 to 0.094. The strength of inference varies across these specifications, and the family-specific adjustment retains its exploratory interpretation.

Table A9. Poverty-gradient sensitivity

	Download speed			Observed tile shares		
Outcome	Mean (log) (1)	P25 (log) (2)	Median (log) (3)	< 10 (pp) (4)	< 25 (pp) (5)	≥ 50 (pp) (6)
A. Excluding Limpopo (198 municipalities)						
Baseline poverty (10 pp)	-0.0872*	0.0427	-0.0094	-2.6997	0.5167	-3.1799**
	(0.0481)	(0.0648)	(0.0480)	(1.0676)	(1.1733)	(1.7753)
Cluster p	0.113	0.531	0.850	0.039	0.673	0.116
Wild-cluster p	0.070	0.750	0.836	0.289	0.836	0.047
BH-adjusted q	0.281	0.836	0.836	0.289	0.836	0.094
Observations	198	198	198	198	198	198
Province clusters	8	8	8	8	8	8
B. Available observations (225 municipalities)						
Baseline poverty (10 pp)	-0.0955**	0.0065	-0.0236	-2.1018	0.3815	-3.5066**
	(0.0452)	(0.0776)	(0.0503)	(1.4563)	(1.0533)	(1.7256)
Cluster p	0.068	0.935	0.651	0.187	0.727	0.077
Wild-cluster p	0.035	0.906	0.652	0.395	0.879	0.023
BH-adjusted q	0.141	0.906	0.906	0.395	0.906	0.047
Observations	225	225	225	225	225	225
Province clusters	9	9	9	9	9	9
C. 2021 vs. 2024 (223 municipalities)						
Baseline poverty (10 pp)	-0.0953	-0.0179	-0.0392	-2.4556	-0.0399	-3.2654**
	(0.0611)	(0.0822)	(0.0483)	(1.9284)	(1.1140)	(1.5772)
Cluster p	0.157	0.833	0.440	0.239	0.972	0.072
Wild-cluster p	0.105	0.852	0.406	0.551	0.984	0.027
BH-adjusted q	0.422	0.984	0.812	0.551	0.984	0.055
Observations	223	223	223	223	223	223
Province clusters	9	9	9	9	9	9
Province FE	Yes	Yes	Yes	Yes	Yes	Yes
Unit of analysis	Municipality	Municipality	Municipality	Municipality	Municipality	Municipality

Notes: The panels contain 198, 225, and 223 municipalities, respectively. Province effects and the ten-percentage-point poverty regressor are retained. The first two panels compare 2021 with 2023–2024; the third compares full-year 2021 with full-year 2024. Stars use the wild-cluster probabilities. BH values retain the same reported two-threshold and four-outcome families within each specification; these are reporting categories rather than documented pre-analysis choices. The exclusion panel has eight province clusters, and the other panels have nine.

The quartile comparison permits a nonlinear relationship with baseline poverty. Municipalities are assigned to fixed groups of 56, 56, 55, and 56 observations, from least to most poor. Figure A5 gives unadjusted levels, whereas Table A10 reports changes relative to the least-poor quartile after province effects. The intermediate groups do not trace a smooth linear gradient. The Q4–Q1 high-speed estimate is imprecise after adjustment.

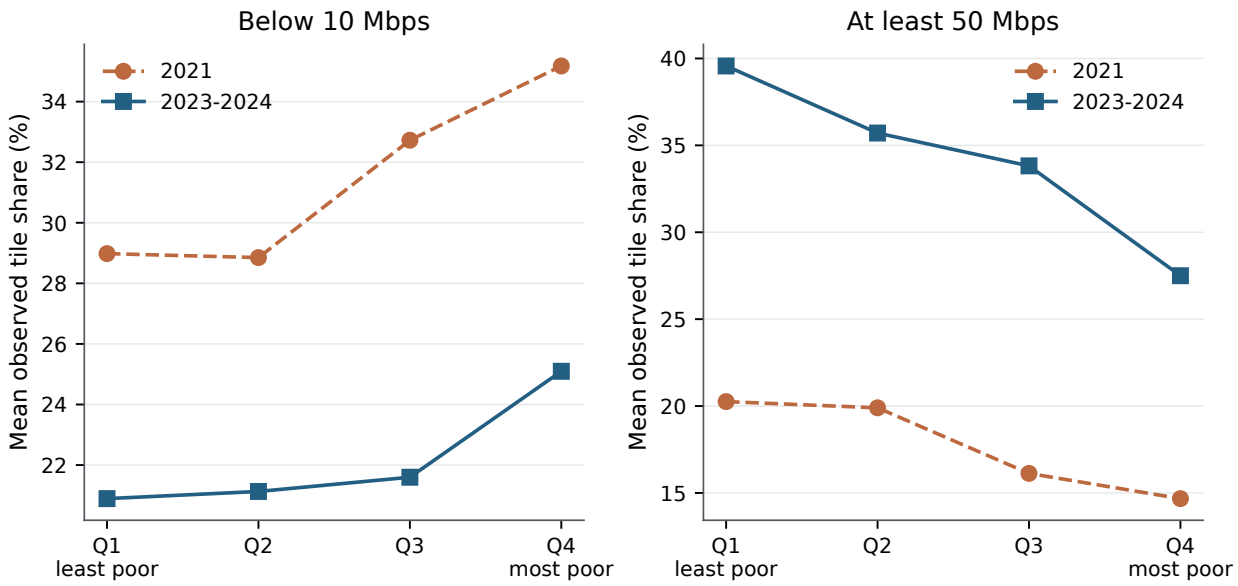


Figure A5. Performance levels by fixed baseline-poverty quartile

Notes: Unadjusted mean municipal tile shares for the same 223 municipalities in 2021 and 2023–2024. Groups use the fixed 2016 poverty-headcount quartiles. Q1 is least poor; Q4 is most poor. Municipalities and quarters receive equal weights within each period. The group levels differ from the province-adjusted changes in Table A10.

Table A10. Province-adjusted changes by baseline-poverty quartile

	Download speed			Observed tile shares		
Outcome	Mean (log)	P25 (log)	Median (log)	< 10 (pp)	< 25 (pp)	≥ 50 (pp)
	(1)	(2)	(3)	(4)	(5)	(6)
Q2 - Q1	-0.0412	-0.0057	-0.0829	0.8621	2.8391	-2.6662*
	(0.0383)	(0.0478)	(0.0460)	(1.2873)	(1.3144)	(1.2866)
Wild-cluster p	0.328	0.906	0.102	0.574	0.141	0.090
BH-adjusted q	0.762	0.957	0.422	0.689	0.422	0.270
Q3 - Q1	0.0288	0.1160	0.0264	-2.9047	-1.2197	-0.0609
	(0.0525)	(0.0593)	(0.0710)	(1.6362)	(1.6247)	(2.0044)
Wild-cluster p	0.582	0.141	0.719	0.191	0.484	0.973
BH-adjusted q	0.776	0.422	0.862	0.383	0.762	0.973
Q4 - Q1	-0.1268*	0.0231	-0.0629	-2.7283	1.6053	-5.1333*
	(0.0616)	(0.0978)	(0.0738)	(1.9546)	(1.7330)	(2.5310)
Wild-cluster p	0.078	0.957	0.508	0.328	0.484	0.059
BH-adjusted q	0.422	0.957	0.762	0.492	0.762	0.270
Province FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	223	223	223	223	223	223
Province clusters	9	9	9	9	9	9

Notes: 223 municipalities. Changes from 2021 to 2023–2024 are regressed on province effects and indicators for Q2, Q3, and Q4, with the least-poor Q1 omitted. Stars use unadjusted wild-cluster p. BH is computed separately over the six contrasts for the two-threshold family and the twelve contrasts for the other four outcomes. Log outcomes and percentage-point share outcomes retain the units used in the main tables.

C.1 Initial-speed control on the poverty geography

This check addresses a specific comparison concern: municipalities with different poverty levels may already have had different service quality before the outcome window. We augment the original poverty change regression with initial mean speed:

$$\Delta Y_i = \widetilde{c}_{p(i)} + \theta^s(H_{i,2016}/10) + \eta\overline{\ell}_{i,2019} + \widetilde{e}_i.$$

Here $\overline{\ell}_{i,2019}$ averages $\log M_{iq}$ across observed 2019 quarters, using the test-weighted municipal mean M_{iq} defined in Section 3. The tile measurements are aggregated on the same 2011 boundaries as the poverty analysis. This control is not standardized. All 223 municipalities have a baseline: 213 have four observed quarters, nine have three, and one has two. We average the available quarters without imputation or sample exclusion. The later outcome change remains the equal-quarter 2023–2024 average minus the equal-quarter 2021 average; 2022 remains excluded.

Table A11 changes only the inclusion of this initial-speed control. Both panels retain equal municipal regression weights, province fixed effects, and the ten-percentage-point poverty regressor. CR1 inference uses nine province clusters and $t(8)$. Restricted wild-cluster tests use all 512 Rademacher sign patterns. BH adjustment covers the six poverty coefficients separately in each panel; it does not cover both specifications jointly or the other analytical choices in the paper. The original six estimates were reproduced before fitting the augmented model. Both models' coefficients and clustered covariance matrices were also checked against a separate implementation.

Table A11. Poverty gradients before and after controlling for 2019 initial speed

	Download speed			Observed tile shares		
Outcome	Mean (log)	P25 (log)	Median (log)	< 10 (pp)	< 25 (pp)	≥ 50 (pp)
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. Original poverty specification						
Baseline poverty (10 pp)	-0.0980	0.0049	-0.0266	-1.9212	0.4651	-3.4798
	(0.0473)	(0.0789)	(0.0517)	(1.4899)	(1.0559)	(1.7128)
CR1 p	0.0719	0.9518	0.6210	0.2332	0.6712	0.0767
Wild-cluster p	0.0352	0.9141	0.6211	0.4492	0.8398	0.0234
BH q (all six)	0.1055	0.9141	0.9141	0.8984	0.9141	0.1055
2019 initial-speed control	No	No	No	No	No	No
Panel B. Add 2019 initial mean speed						
Baseline poverty (10 pp)	-0.0952	0.0061	-0.0372	-1.3417	1.1802	-3.2291
	(0.0424)	(0.0781)	(0.0581)	(1.5995)	(1.1609)	(1.4355)
CR1 p	0.0550	0.9400	0.5398	0.4259	0.3391	0.0546
Wild-cluster p	0.0313	0.9297	0.5586	0.6328	0.3164	0.0117
BH q (all six)	0.0938	0.9297	0.7594	0.7594	0.6328	0.0703
2019 initial-speed control	Yes	Yes	Yes	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	223	223	223	223	223	223
Municipalities	223	223	223	223	223	223
Province clusters	9	9	9	9	9	9

Notes: The same 223 municipalities on 2011 boundaries enter both panels. Outcomes are changes from 2021 to 2023–2024; logs are averaged before differencing. Poverty coefficients are per ten percentage points of the 2016 household poverty headcount. Shares refer to observed tiles, in percentage points. Parentheses contain province-cluster CR1 standard errors. CR1 p uses $t(8)$; wild p enumerates the 512 restricted-null sign patterns. BH q adjusts six poverty coefficients within each panel, using the same family for both specifications. No stars are used. The initial-speed control averages quarterly log test-weighted mean speeds in observed 2019 quarters. Panel A reproduces Table 3; its earlier 2+4 family adjustment is not repeated here.

The controlled high-speed coefficient has a CR1 95 percent interval of $[-6.5393, 0.0812]$ percentage points. The mean-log-speed interval is $[-0.1930, 0.0026]$. The wild probabilities are below 0.05, but both six-outcome adjusted values exceed 0.05. Thus this exercise preserves the negative pattern while leaving its statistical strength dependent on the inference convention. Conditioning on initial mean speed does not hold devices, users, or test selection fixed, and does not identify a mechanism or a causal effect of poverty.

D Measurement Composition and Aggregate Distribution Checks

Figure A6 compares the aggregate tile-speed distribution with and without location turnover. Each tile receives equal weight; this differs from the equal-municipality mean in main Figure 1. The fixed-location sample is selected, as documented below.

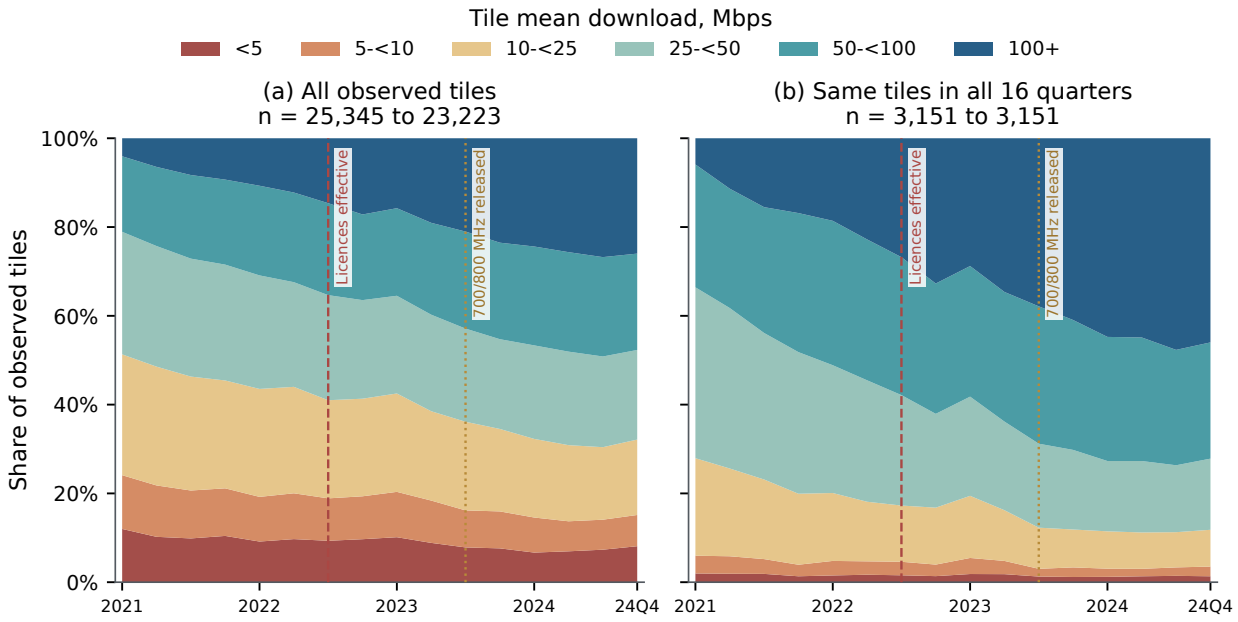


Figure A6. The distribution of observed tiles across download-speed bands

Notes: Quarterly tile distributions, 2021Q1–2024Q4. Each tile receives equal weight within a quarter; the six bands classify its mean download speed. Panel (a) includes all observed tiles. Panel (b) includes the same 3,151 tiles in every quarter. The boundaries are < 5, $5 \leq v < 10$, $10 \leq v < 25$, $25 \leq v < 50$, $50 \leq v < 100$, and $v \geq 100$ Mbps. The samples have different socioeconomic composition. Vertical labels mark the licence-validity quarter (2022Q3) and quarter when the 700/800 MHz bands were freed for mobile (2023Q3).

D.1 Measurement intensity and fixed-location support

Table A12 uses the unchanged 223-municipality poverty sample. Panel A replaces the outcome in Equation (4) with the change in three log measurement-intensity measures. It is a diagnostic regression, not an extra control in the main poverty equation. Panel B shows which municipalities contain at least one tile observed in all sixteen quarters. Its poverty quartiles are fixed from the same baseline headcounts as Appendix C.

The selected fixed-location regression uses the same pre/post averaging and province effects, but only municipalities with continuous support. Its high-speed coefficient is -7.7502 (SE 9.3156), with CR1 $p = 0.430$, wild $p = 0.516$, and a conventional 95 percent interval of $[-29.2320, 13.7316]$ percentage points. The sample contains all nine provinces but represents only 117 of the 223 municipalities. Its wider interval and uneven poverty coverage prevent a population-wide interpretation.

Table A12. Poverty, measurement intensity, and continuous-location support

A. Poverty and changes in log measurement intensity

Outcome	Coefficient	CR1 SE	CR1 p	Wild p	BH q
Log tiles	0.080	0.080	0.347	0.480	0.480
Log tests	0.125	0.098	0.235	0.422	0.480
Log tests per tile	0.046	0.021	0.060	0.199	0.480

B. Continuous-location support by fixed poverty quartile

Quartile	Main places	Supported	Continuous tiles	Supported share (%)
Q1 least poor	56	39	2,294	69.6
Q2	56	37	625	66.1
Q3	55	30	207	54.5
Q4 poorest	56	11	25	19.6

Notes: Panel A compares the mean log measure in 2023–2024 with its 2021 mean, with province effects and the headcount regressor in units of ten percentage points. There are 223 municipalities and nine province clusters. CR1 p uses $t(8)$; wild p enumerates 512 sign patterns; BH covers the three diagnostics. Panel B counts tiles observed in every quarter, within the original poverty-sample membership. The supported subset has 117 municipalities; it is not the primary sample.

The continuously observed tiles are concentrated in low-poverty municipalities and form a small subset of the full quarterly footprint (Figure A7). They show performance changes at fixed locations. Comparing their path with the full sample helps assess the role of changing measurement locations.

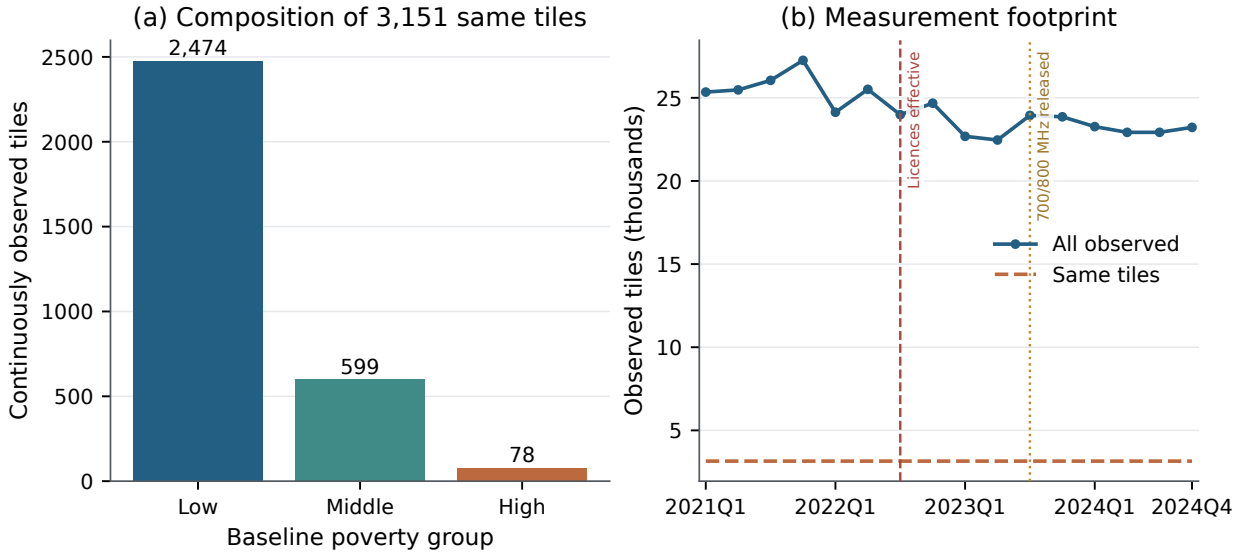


Figure A7. Continuous observation and the measurement footprint

Notes: Panel (a) counts the 3,151 continuously observed tiles by the 2016 poverty tercile classification: 2,474 low-poverty, 599 middle-poverty, and 78 high-poverty. These terciles are distinct from the poverty quartiles in Appendix C. Panel (b) compares all observed tile counts with the same-tile count each quarter.

E GDP-scaled Nighttime Lights

This appendix reports an auxiliary GDP-scaled nighttime-light outcome comparison under the same policy specification used for internet performance. It brings together the data, the raw and background-masked light outcomes, and the construction and trend diagnostics. Treatment classifications, sample, time window, fixed effects, and controls are retained. The exercise supplies an additional policy-comparison outcome. The main paper studies the distribution of measured service-quality improvements.

E.1 Nighttime lights and national GDP for the activity extension

The economic-activity extension uses quarterly sums of VIIRS nighttime radiance on the same 2018 municipal boundaries. The underlying monthly product is the Earth Observation Group's stray-light-corrected, cloud-free VIIRS composite, accessed through the VCMSLCFG collection.[8] For each native-grid pixel, we average the available monthly radiances within the calendar quarter. Months receive equal weight. Observations with zero cloud-free coverage are excluded. We then sum valid quarterly pixel values whose centers fall inside each municipality. Valid zero and negative pixel radiances are retained in the raw extract. Municipal sums are positive throughout the analysis window. This statistic is a sum of pixel radiances, rather than an area-weighted integral of emitted power.

We retain both the raw sum and a background-masked sum. The latter applies the provider's annual lit-area support to each quarter of the corresponding year. It uses V21 for 2019–2021 and V22 for 2022–2024. Elvidge et al. (2021) describe the annual VIIRS processing approach.[9] Our quarterly construction uses the annual spatial support while retaining the quarterly radiances inside it. It therefore does not reproduce all temporal filtering of the provider's annual product. Subsections E.5 and E.6 give the construction and diagnostic details.

National GDP is production-approach GDP at market prices from Statistics South Africa, expressed in million rand at constant 2015 prices. We use seasonally adjusted, nonannualised quarterly amounts from the 2026Q2 release vintage. We obtain these amounts by dividing the official annualised series by four.[10] The construction covers 2019Q1–2024Q4. We first allocate each quarter's national GDP across all 213 municipalities. We then restrict the regression to the original 206 places. It yields a model-based proxy, rather than an official

[8] https://developers.google.com/earth-engine/datasets/catalog/NOAA_VIIRS_DNB_MONTHLY_V1_VCMSLCFG. The radiance band is avg_rad; cf_cvg records cloud-free observation coverage.

[9] The EOG annual and monthly product definitions, including the annual lit masks, are available at https://eogdata.mines.edu/products/vnl/.

[10] Statistics South Africa, P0441, national accounts time series: https://www.statssa.gov.za/?PPN=P0441&page_id=1854. The source seasonally adjusted series is QRS1000; dividing its level by four changes its reporting scale, not its seasonal adjustment.

municipal GDP series. The sum measure targets total local activity and is not divided by population.

E.2 From light sums to a GDP-scaled proxy

Nighttime lights can supplement economic measurement where local output statistics are unavailable. Henderson, Storeygard, and Weil (2012) develop a measurement-error framework combining lights and official growth statistics. A related spatial-allocation approach uses national GDP-to-light ratios to map economic output within countries (Sutton and Costanza, 2002). Ghosh et al. (2010) extend spatial allocation with population information for agricultural activity. Their approach illustrates why light alone may miss output from dimly lit sectors. The direct precedent for using lights as an internet-related outcome is Goldbeck and Lindlacher (2025). Their annual town-level internet-access design differs from our quarterly municipal mobile-performance comparison.

Let $S_{it}^{(m)}$ denote the municipal sum in quarter t, where m identifies raw or background-masked lights. Let G_t denote national real quarterly GDP. We construct

$$w_{it}^{(m)} = \frac{S_{it}^{(m)}}{\sum_{j=1}^{213} S_{jt}^{(m)}}, \qquad \widetilde{G}_{it}^{(m)} = G_t w_{it}^{(m)}. \tag{6}$$

The allocation shares sum to one, and the municipal proxies sum to national GDP each quarter. The common conversion factor, $G_t / \sum_j S_{jt}^{(m)}$, is updated quarterly. It is not an independently estimated municipal light-to-output coefficient. Raw and masked proxies use their own full-universe denominators.

This proportional allocation imposes a common light-to-output ratio within each quarter. It does not estimate that relationship from municipal accounts. Gibson et al. (2021) compare lights and subnational GDP in Indonesia, China, and South Africa. VIIRS performs better than DMSP, but difficulties remain in low-density areas and primary-sector activity. Gibson, Kim, and Li (2024) show that growth elasticities vary with spatial scale, metropolitan status, period, and sensor. Beyer, Hu, and Yao (2022) develop a quarterly measurement framework that uses information on effective light observations. It does not establish a universal municipality conversion factor. Recent validation by Chiovelli et al. (2026) also distinguishes cross-sectional and over-time relationships and shows the importance of consistent processing. These findings motivate interpreting $\widetilde{G}_{it}^{(m)}$ as a GDP-scaled light proxy and checking its temporal behavior.

E.3 The original policy comparison with a different outcome

We replace the outcome in Equation (5) with the log proxy:

$$\log \widetilde{G}_{ipt}^{(m)} = \alpha_i + \lambda_{pt} + \mu_{B_i,t} + \beta_m(Early_p \times B_i \times Post_t) + X_{it}'\gamma_m + \varepsilon_{ipt}^{(m)}. \tag{7}$$

Here B_i is the unchanged ICASA Batch 3 indicator, rather than a poverty classification. As in the internet regression, the baseline is 2021, the post period is 2023–2024, and 2022 is excluded. The sample remains 206 municipalities and 2,472 equally weighted municipality-quarter observations. Municipality, province-by-quarter, and Batch-3-status-by-quarter effects are retained. The controls remain log valid tile count and observed tile footprint share. Province-cluster CR1 inference uses $t(8)$. For this extension we additionally report restricted wild-cluster p-values. These use all 512 Rademacher sign patterns across the nine provinces.

The coefficient compares the Batch 3–other-municipality difference in proxy changes between Early and Late provinces. It conditions on the original controls. An important property of the construction is

$$\log \widetilde{G}_{it}^{(m)} = \log S_{it}^{(m)} + \log G_t - \log\left(\sum_{j=1}^{213} S_{jt}^{(m)}\right). \tag{8}$$

The last two terms are common to all municipalities in a quarter. Province-by-quarter effects absorb them. Thus the DDD coefficient and standard error equal those from log light sums under the same specification. Scaling to GDP units does not supply an independent source of local economic variation.

Table A13 places the reproduced mean-speed result next to the two activity outcomes. The background-masked sum is the cleaned measurement; the raw sum is its benchmark.

Table A13. The original policy DDD with internet and economic-activity outcomes

Outcome	Mean speed (log) (1)	Raw Sum GDP proxy (log) (2)	Masked Sum GDP proxy (log) (3)
Early × Batch 3 × Post	0.1998	-0.0241	-0.1106
	(0.0653)	(0.0303)	(0.0580)
CR1 p (unadjusted)	0.016	0.449	0.093
Wild-cluster p	0.043	0.484	0.113
CR1 95% interval	[0.0492, 0.3505]	[-0.0939, 0.0457]	[-0.2443, 0.0232]
Municipality FE	Yes	Yes	Yes
Province × Quarter FE	Yes	Yes	Yes
Batch 3 status × Quarter FE	Yes	Yes	Yes
Log valid tile count	Yes	Yes	Yes
Observed tile footprint share	Yes	Yes	Yes
Differential linear trend	No	No	No
Observations	2,472	2,472	2,472
Municipalities	206	206	206
Quarters	12	12	12
Province clusters	9	9	9

Notes: Column (1) reproduces the mean-speed specification in Table A4. Columns (2)–(3) change only the dependent variable to the log real, seasonally adjusted, nonannualised GDP proxy allocated by raw or background-masked Sum. The allocation denominator contains all 213 municipalities; the regression contains the same 206 municipalities in 2021 and 2023–2024, excluding 2022. Parentheses contain province-cluster CR1 standard errors. Conventional p-values and 95 percent intervals use $t(8)$. Wild-cluster p-values use the restricted-null bootstrap and all 512 sign patterns. Probabilities are unadjusted for multiple testing. We report probabilities explicitly, following the policy-table convention.

The raw-sum coefficient is -0.0241 (SE 0.0303), with conventional $p = 0.449$ and wild-cluster $p = 0.484$. The background-masked coefficient is -0.1106 (SE 0.0580), with conventional $p = 0.093$ and wild-cluster $p = 0.113$. Its point estimate corresponds to a 10.5 percent negative relative contrast, using $100[\exp(\beta_m) - 1]$. However, its conventional 95 percent log-point interval, $[-0.2443, 0.0232]$, includes zero. The evidence therefore does not clearly distinguish the masked estimate from zero under the additional small-cluster test.

The activity outcomes also require their own trend assessment. A four-quarter 2021 linear diagnostic rejects a zero differential trend for raw Sum under the wild test ($p = 0.020$). The masked diagnostic does not ($p = 0.344$). A separate longer-window diagnostic extending the same classifications and fixed effects to 2019–2024 rejects the masked series' joint 2019–2021 coefficients ($p = 0.016$). Subsection E.6 reports both exercises and their different samples. These patterns further limit a causal interpretation of the activity extension. Non-rejection in the shorter diagnostic does not validate parallel trends (Roth, 2022).

E.4 What the extension adds

The original policy comparison yields a positive internet-performance contrast. Over this horizon, the light-based activity proxy does not show clear evidence of a positive contrast. The negative masked point estimate is too weak and trend-sensitive to support a conclusion that internet improvement reduced local GDP. Nor does imprecision establish that economic activity was unaffected. Lights respond to lighting conditions and the sectoral composition of activity, and their conversion to GDP remains model-based.

The internet and activity regressions are separate conditional policy contrasts. They do not identify the effect of a one-percent increase in measured speed on economic activity. Nor do they establish that speed mediates the policy contrast. That interpretation would require additional evidence and identifying assumptions. The extension therefore complements the service-quality and poverty findings. The existing analyses establish the distribution of measured upgrading. This additional proxy outcome does not establish an economic-activity increase.

E.5 Spatial and temporal construction

We use the same unsimplified 2018 municipal polygons throughout. Native VIIRS pixel centers are included when they fall strictly inside a municipality. Pixel values are not resampled to a new grid. Available monthly radiances are averaged within each calendar quarter, and the quarterly pixel radiances are summed. Zero cloud-free coverage denotes an unavailable monthly value, which is excluded from that average. Observed zero and negative radiances are retained. Availability can therefore affect the sum as well as brightness. The resulting raw and background-masked aggregate sums are positive for all 213 municipalities in all 24 quarters.

For masked Sum, the annual EOG lit support of the corresponding year is applied to the quarterly native grid. The radiances inside this support are retained. The mask does not itself remove temporary lights within retained pixels. Year-specific support permits newly lit areas to enter and may vary with local conditions. The V21/V22 product versions are recorded as part of the measurement construction. Annual products and averages of monthly composites need not have interchangeable growth rates (Beyer, Hu, and Yao, 2022). The raw and masked measures are consequently reported separately.

The municipal GDP proxy is constructed over all 213 places, with 5,112 municipality-quarter records in 2019Q1–2024Q4. The denominator is not recalculated after selecting the 206-region policy sample. National real GDP is seasonally adjusted and expressed as a three-month amount in constant-2015 million rand. Population does not enter the conversion, because the target is total local activity rather than activity per person. This choice does not

remove the influence of settlement patterns, lighting technology, or economic composition on the measure.

E.6 Trend diagnostics

Panel A of Table A14 uses only 2021Q1–Q4. It replaces the post indicator in the original DDD interaction with a linear quarter index. The original fixed effects and two controls are retained. The displayed coefficient is the differential log-proxy trend per quarter. This diagnostic has 824 observations in the same 206 municipalities.

Panel B extends the observation window to 2019Q1–2024Q4 with unchanged municipality membership, treatment labels, fixed-effect families, and controls. It uses the 4,930 municipality-quarter cells jointly observed for internet and the proxies. Fourteen 2019–2020 internet observations are unavailable. Quarter-specific $Early \times Batch3$ interactions are normalised to 2021Q4. The test covers the eleven 2019–2021 coefficients other than that omitted quarter. Its statistic is the largest absolute studentised coefficient. We evaluate it with a restricted-null wild-cluster bootstrap over 512 sign patterns. This is a supplementary trend diagnostic, not the sample used for the main scalar result.

Table A14. Pre-period diagnostics for the activity outcomes

Outcome	Raw Sum proxy (log)	Masked Sum proxy (log)
A. Linear differential trend within 2021		
Early × Batch 3 × quarter index	-0.0251	-0.0101
	(0.0099)	(0.0120)
CR1 p	0.035	0.424
Wild-cluster p	0.020	0.344
Observations	824	824
Municipalities	206	206
Quarters in diagnostic	4	4
B. Joint pre-period test in the 2019–2024 panel		
Maximum absolute t	3.451	8.413
Wild joint p	0.098	0.016
Pre-period restrictions	11	11
Panel observations	4,930	4,930
Municipalities	206	206
Province clusters	9	9

Notes: Panel A retains the original 2021 sample and reports the coefficient on $Early \times Batch3\times$ a linear quarter index, its province CR1 standard error, and conventional $t(8)$ and restricted wild-cluster probabilities. Panel B reports a joint maximum-absolute-t wild test of eleven quarter interactions in 2019–2021 relative to 2021Q4, from the longer 2019–2024 panel. Both panels retain municipality, province-by-quarter, and Batch-3-status-by-quarter fixed effects and the original two controls. Nine province clusters are used. Probabilities are unadjusted. The two panels have different windows and different hypotheses.

F Comparison with Nearby Literature

The broader connectivity literature includes employment responses to international connectivity in Africa (Hjort and Poulsen, 2019) and household welfare responses to mobile coverage in Nigeria (Bahia et al., 2024). The upgrade studies discussed in the Introduction address a distinct performance margin. The comparisons below also retain technology geography in Australia (Alizadeh et al., 2024), mobile usage and mobility in Spain (Herrero-Zamorano et al., 2026), and the socioeconomic measurement preprint by Kalamadi et al. (2026).

Tables A15–A16 compare eight nearby studies. OECD (2025) provides an additional policy benchmark. Its fixed-network contrasts use relative means and absolute medians (Figures 1.2 and 2.6). Its mobile benchmarks use Ookla and Opensignal (Figures 2.10 and 2.14). These geographic classifications differ from ours.

Table A15. Nearby studies: speed inequality, temporal change, and technology geography

Study	Data and comparison	Relation to this paper
Riddlesden and Singleton (2014)	Fixed broadband. England; tests/postcodes and contextual area classifications. 2010-2013; short-run time-of-day and multi-year comparisons. Descriptive spatial and temporal analysis.	Rising average speeds coexist with geographic and social disparities; peak-hour and rurality patterns differ. A direct temporal precedent. Our addition is quarterly mobile upgrading in South Africa, fixed 2019 groups, separate absolute/relative/threshold margins, and a matched poverty comparison.
Hilbert (2016)	Installed communications bandwidth. 172 countries. 1986-2014. Longitudinal inequality and technology-diffusion comparison.	Bandwidth inequality evolves with technological progress and diffusion; counting connections misses capacity differences. We study realized mobile test performance within one country, rather than installed national capacity.
Gallardo and Whitacre (2024)	Fixed broadband. 82,519 continental U.S. census tracts. 2021 tests; ACS 2017-2021. Cross-sectional spatial regressions of annual download/upload speeds.	Several adoption disadvantages also appear in measured speeds; the racial-composition pattern differs from a simple adoption narrative. We follow changes in quarterly mobile outcomes and distinguish initial quality from household poverty on historically consistent municipal frames.
Alizadeh et al. (2024)	Mixed NBN technologies. Australia; technology footprint linked to ABS/remoteness geography. July 2020 footprint; discussion of 2020-2022 upgrade rounds. Geospatial equity analysis and policy assessment.	The technology footprint exhibits an urban-regional divide; promised upgrades raise further distributional questions. We measure realized quarterly performance, including low-speed and high-speed tile shares, rather than treating technology availability as delivered speed.

Notes: Temporal and multidimensional digital-divide analysis predates this paper. The present contribution concerns the joint description of distinct mobile-performance margins during South Africa's spectrum transition. Alizadeh et al. appeared online in 2023 and in the 2024 issue. Full references and persistent links appear in the bibliography.

Table A16. Nearby studies: measurement, mobile use, and socioeconomic performance

Study	Data and comparison	Relation to this paper
Bogulski et al. (2024)	M-Lab and Ookla fixed/mobile. U.S. counties. 2020-2021. Negative-binomial models with robust standard errors.	Rural underrepresentation appears in fixed-test counts; the reported rurality association is absent for mobile counts. We test poverty-related measurement-intensity changes and continuous-tile support in the actual South African mobile sample.
Frias, Lehr and Stocker (2025)	Mobile broadband measurement. Measurement ecosystem; no single municipal estimation sample. Conceptual/policy synthesis, published 2025. Measurement-method and governance analysis.	Measurement methods have context-dependent trade-offs; policy questions require appropriate data and governance. We implement a transparent distributional monitoring application and report its coverage constraints.
Herrero-Zamorano et al. (2026)	Mobile voice/data traffic and fixed wireless access. Spain; 6,081 of 8,116 municipalities. A full year; August versus November and weekday/weekend contrasts; calendar year not specified in inspected data section. Linear and Random Forest prediction across municipality geotypes.	Population alone predicts demand poorly, especially in small municipalities; visitors, seasonality and FWA matter. We study delivered mobile speed and its distribution, not traffic-demand prediction or a resident-population model.
Kalamadi et al. (2026) Preprint	Fixed broadband. Six U.S. metropolitan regions; census block groups. 2021-2025. Sampling reweighting, density stratification, Random Forest/permutation importance.	Socioeconomic predictors vary across density strata and regions; sampling correction is part of the framework. Our question concerns quarterly mobile change and distinct performance margins during a spectrum transition.

Notes: Bogulski et al.'s fixed/mobile distinction is retained. Frías et al. is a measurement-policy study. Herrero-Zamorano et al. predicts traffic rather than speed. Kalamadi et al. is cited as the May 2026 arXiv preprint; peer-reviewed publication is not established here. These studies do not identify the source of our poverty gradient.